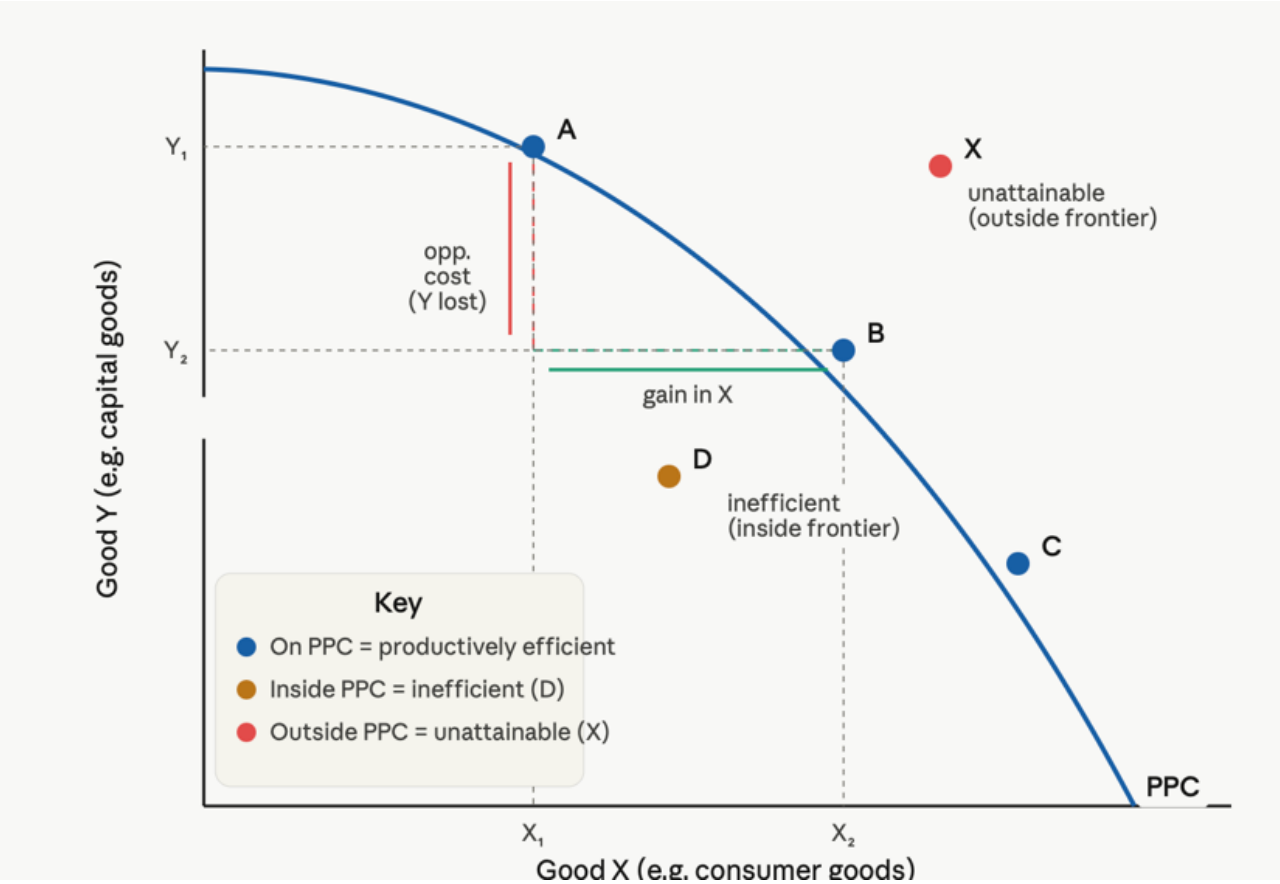
PAPER 1 MICRO AUGUST (DP-1)

Q1 Explain how the concepts of scarcity, choice, and opportunity cost are interrelated in the allocation of resources. [10]
Ans Scarcity, choice and opportunity cost are the three foundational concepts of economics, and they are so deeply interrelated that each one logically implies the other two. Together they explain why the problem of resource allocation exists and why every economic decision involves a trade-off.

Scarcity is the starting point. It refers to the fundamental condition in which human wants are unlimited while the resources available to satisfy those wants — land, labour, capital and entrepreneurship — are finite. This is not a temporary or correctable problem but a permanent feature of the human condition. Even the wealthiest economies in the world cannot produce everything their populations desire, and even individuals with substantial incomes cannot satisfy every want simultaneously. Scarcity applies universally — to households, firms, and governments alike — and it is the existence of scarcity that makes economics a necessary discipline.

Because resources are scarce, they cannot be used for all purposes simultaneously. A plot of land cannot be both a hospital and a housing estate at the same time. A government budget cannot fund both a tax cut and an increase in public spending without limit. Scarcity therefore makes choice inevitable — every economic agent must decide how to allocate the limited resources at their disposal among competing uses. Without scarcity there would be no need to choose, and without choice the concept of resource allocation would be meaningless.

It is precisely because choices must be made that opportunity cost arises. Opportunity cost is defined as the value of the next best alternative forgone when a choice is made. When a government chooses to allocate scarce tax revenue to building new roads, the opportunity cost is whatever else that money could have produced — perhaps new schools or hospitals. Opportunity cost is not the monetary cost of the chosen option but the real cost measured in terms of the alternatives sacrificed. The production possibility curve below illustrates all three concepts simultaneously:



The PPC diagram above shows the maximum combinations of Good X (consumer goods) and Good Y (capital goods) that an economy can produce using all its scarce resources efficiently. Every point on the curve represents a choice about how to allocate those resources, and every movement along the curve makes the opportunity cost explicit.

Consider the movement from point A to point B. At A, the economy produces a large quantity of Good Y (Y■) but relatively little of Good X (X■). If the economy chooses to produce more of Good X — moving to point B where output is X■ — it must release resources from the production of Good Y. Output of Y falls from Y■ to Y■. This loss of Y is the opportunity cost of producing more X. The red arrow shows what is sacrificed; the green arrow shows what is gained. Crucially, the two arrows are not equal in length — the PPC is concave to the origin, reflecting the law of increasing opportunity cost. As more and more resources are shifted from Y to X, those resources become progressively less suited to producing X, meaning ever-larger quantities of Y must be sacrificed for each additional unit of X gained. This is why the curve bows outward rather than being a straight line.

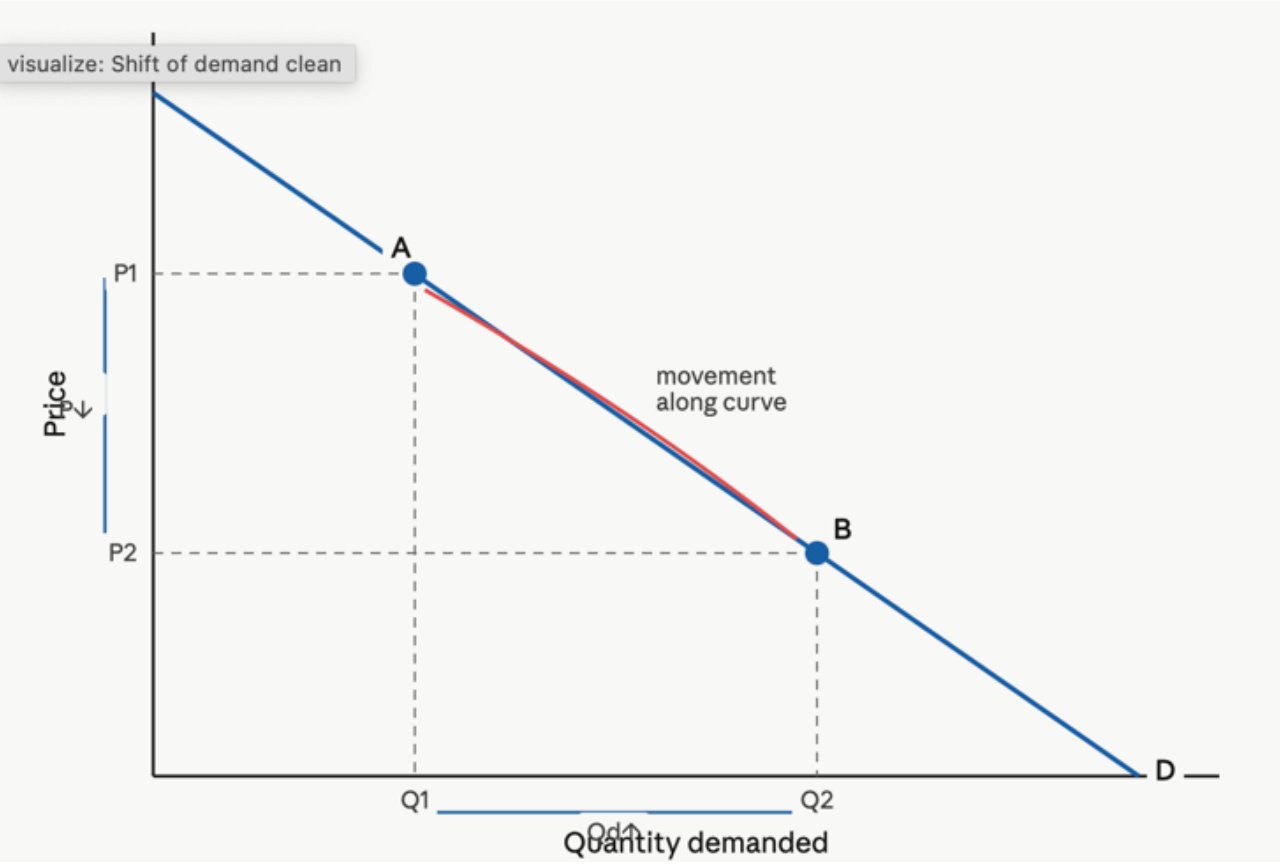
Point D lies inside the frontier. This represents a situation where the economy is not making full or efficient use of its scarce resources — some factors of production are idle, misallocated or unemployed. At D, more of both goods could be produced by moving to the frontier, which means no opportunity cost need be incurred to expand output. This might occur during a recession when unemployment is high. Point X lies outside the frontier entirely — it is unattainable given the economy's current resource endowment and technology. To reach X, the economy would need either an increase in the quantity of resources (for example through population growth or capital accumulation) or an improvement in technology, both of which would shift the entire PPC outward.

The three concepts are therefore logically inseparable and all visible in the diagram. Scarcity is represented by the frontier itself — the boundary that separates what is attainable from what is not. Choice is represented by the selection of any particular point on the curve — the economy must decide how to divide its scarce resources between Good X and Good Y. Opportunity cost is represented by the slope of the curve at any point — it is the rate at which one good must be sacrificed to produce more of the other, and it is positive and increasing precisely because resources are scarce and not perfectly adaptable between uses.

In conclusion, scarcity, choice and opportunity cost are interrelated in the following way: scarcity is the condition that makes resource allocation necessary, choice is the mechanism through which that allocation is made, and opportunity cost is the measure of what is sacrificed by every allocation decision. The PPC gives all three concepts a single unified visual representation — the frontier embodies scarcity, every point on it embodies a choice, and its slope at any point embodies the opportunity cost of that choice. None of the three is meaningful without the others, and together they define the central economic problem that every society must solve.

Q2 Explain the difference between movements along the demand curve and shifts of the demand curve. [10]
Ans The demand curve is a graphical representation of the relationship between the price of a good and the quantity demanded, ceteris paribus — all other factors held constant. Understanding the distinction between a movement along the demand curve and a shift of the demand curve is fundamental to demand analysis, as the two phenomena have entirely different causes and entirely different implications for market outcomes.

A movement along the demand curve occurs when the quantity demanded changes solely as a result of a change in the price of the good itself. Because the demand curve plots price against quantity demanded, a change in price moves the consumer to a different point on the same curve — either upward along it if price rises, or downward along it if price falls. The curve itself does not move. This is also referred to as a change in quantity demanded rather than a change in demand. The law of demand states that, ceteris paribus, there is an inverse relationship between price and quantity demanded — as price rises, quantity demanded falls, and vice versa. This is explained by the substitution effect, whereby a higher price makes the good relatively more expensive compared to substitutes, causing consumers to switch away from it, and the income effect, whereby a higher price reduces consumers' real purchasing power, further reducing consumption.



As shown in diagram 1, when price falls from P1 to P2, the consumer moves from point A to point B along the same demand curve D. Quantity demanded increases from Q1 to Q2. The curve itself is unchanged — only the point on it has moved. This is a contraction of demand when price rises (A to B in reverse) and an extension of demand when price falls (A to B as shown). In both cases the cause is exclusively a change in the price of the good itself, and no other factor has changed.

A shift of the demand curve, by contrast, occurs when a factor other than the price of the good changes, causing consumers to demand a different quantity at every price level. The entire curve moves — either rightward, indicating an increase in demand, or leftward, indicating a decrease in demand. This is referred to as a change in demand rather than a change in quantity demanded. The non-price determinants that cause shifts include changes in consumer income, changes in the prices of related goods (substitutes and complements), changes in tastes and preferences, changes in population size or demographics, and changes in consumer expectations about future prices.

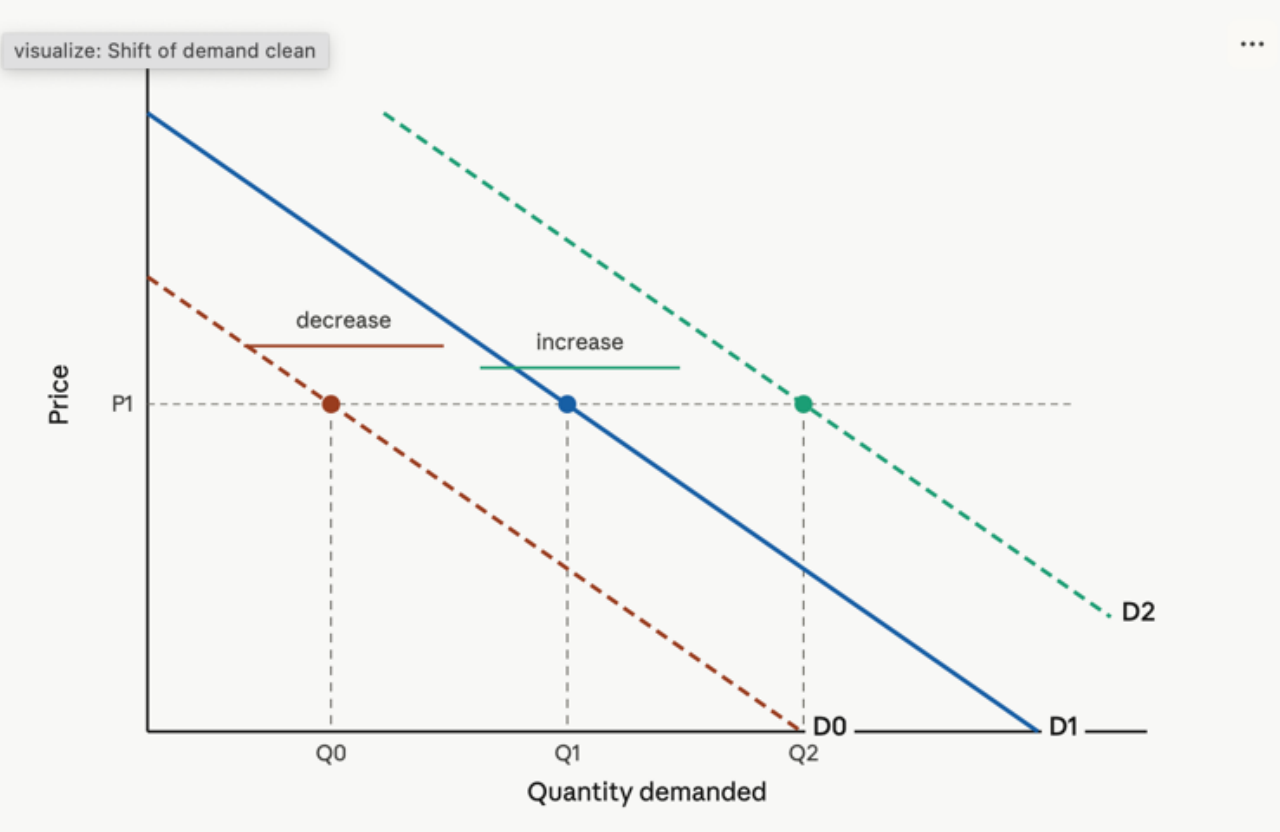


Diagram 2 shows all three curves with the same slope — D0, D1 and D2 are perfectly parallel, confirming that the shift is purely horizontal with no change in the responsiveness of consumers to price. At the same price P1, quantity demanded differs across the three curves: Q0 on D0, Q1 on D1 and Q2 on D2. This illustrates the defining feature of a shift — a different quantity is demanded at every price level, not just at one point.

Consider each determinant in turn. If consumer income rises and the good is a normal good — one for which demand increases with income — consumers can afford to buy more at every price, shifting the curve rightward from D1 to D2. If income falls, or if the good is an inferior good, the curve shifts leftward to D0. If the price of a substitute good rises — for example, if the price of butter rises, increasing demand for margarine — consumers switch toward the good in question, shifting its demand curve rightward. Conversely, if the price of a complement rises — for example, if the price of petrol rises, reducing demand for cars — the demand curve for the complementary good shifts leftward. Changes in tastes and preferences, perhaps driven by advertising or health campaigns, shift the curve in the direction of the change. A government health campaign warning against sugar consumption would shift the demand curve for sugary drinks leftward, while a successful advertising campaign would shift it rightward.

The critical distinction between the two concepts can be summarised as follows. A movement along the demand curve is caused exclusively by a change in the price of the good itself, results in a change in quantity demanded, and leaves the curve in its original position. A shift of the demand curve is caused by a change in any non-price determinant of demand, results in a change in demand, and moves the entire curve to a new position. Confusing the two is one of the most common errors in demand analysis — a firm that mistakes a movement along its demand curve for a genuine shift in consumer preferences, or vice versa, will draw entirely incorrect conclusions

about the strength of demand for its product and make poor pricing and production decisions as a result.

In conclusion, movements along the demand curve and shifts of the demand curve are fundamentally different phenomena with different causes, different graphical representations and different implications. The former reflects the law of demand operating as price changes; the latter reflects changes in the underlying conditions of demand. Both are essential tools for analysing how markets respond to changing economic circumstances.

Q3 Explain how non-price determinants of supply can lead to a shift of the supply curve. [10]
Ans The supply curve shows the relationship between the price of a good and the quantity that producers are willing and able to supply, ceteris paribus. A shift of the supply curve occurs when a factor other than the price of the good itself changes, causing producers to supply a different quantity at every price level. These factors are known as the non-price determinants of supply, and a change in any one of them will shift the entire supply curve either rightward — representing an increase in supply — or leftward — representing a decrease in supply.

The first and most significant non-price determinant is the cost of factors of production. Firms use land, labour, capital and entrepreneurship to produce goods and services, and the prices of these inputs directly affect the cost of production. If input costs rise — for example, if wages increase following a national minimum wage rise, or if raw material prices rise due to a supply disruption — the cost of producing each unit increases. At any given price, producers find it less profitable to supply the same quantity as before and will reduce output. The supply curve therefore shifts leftward. Conversely, if input costs fall — perhaps due to a fall in oil prices reducing transport and energy costs — production becomes cheaper, profitability rises at every price level, and producers are willing to supply more. The supply curve shifts rightward. This determinant is arguably the most powerful driver of supply shifts in practice, as input cost changes affect virtually every industry simultaneously.

A second determinant is the state of technology. An improvement in production technology allows firms to produce a given quantity of output using fewer inputs, thereby reducing the average cost of production. This raises profitability at every price level and gives producers both the ability and incentive to supply more. The supply curve shifts rightward. The development of automation in manufacturing, the adoption of more efficient agricultural techniques, or the introduction of software that streamlines logistics are all examples of technological improvements that have historically shifted supply curves rightward across entire industries. Technology rarely shifts supply curves leftward in practice, as productive knowledge is not typically lost, though obsolescence of capital equipment could temporarily reduce productive capacity.

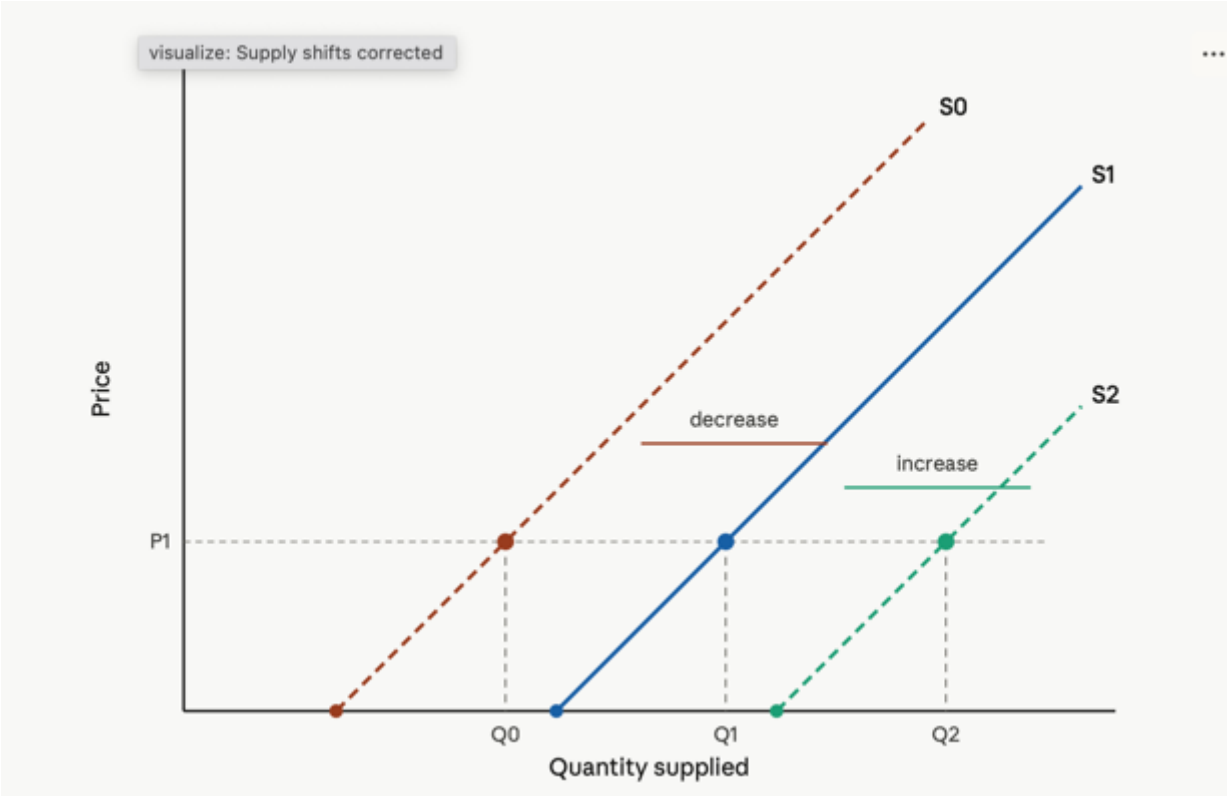
A third determinant is the number of producers in the market. If new firms enter the industry — attracted by profit opportunities or lower barriers to entry — total market supply increases at every price level, shifting the supply curve rightward. If firms exit — due to sustained losses, rising costs or tightening regulation — market supply decreases and the curve shifts leftward. This determinant is particularly relevant in the long run, when entry and exit are possible, and explains why competitive markets tend toward normal profit over time as the supply curve adjusts through firm entry and exit.

A fourth determinant is government intervention in the form of taxes and subsidies. An indirect tax imposed on producers raises their cost of production by the amount of the tax per unit, shifting the supply curve leftward by the vertical distance of the tax. At every quantity, producers now require a higher price to cover their increased costs. A subsidy, by contrast, reduces the effective cost of production, shifting the supply curve rightward as producers can profitably supply more at every price. This distinction is directly relevant to government policy — subsidies on renewable energy production, for example, have shifted the supply curves of solar and wind electricity rightward, increasing output and reducing prices.

A fifth determinant is the prices of related goods in production — goods that use the same factors of production as the good in question. If a farmer can grow either wheat or barley on the same land, and the price of barley rises substantially, the farmer has an incentive to reallocate land and labour toward barley production and away from wheat. The supply curve for wheat therefore shifts leftward even though the price of wheat itself has not changed. This substitution in production is the supply-side equivalent of the substitution effect in consumption and explains how price signals in one market can shift supply curves in related markets.

Finally, expectations about future prices can cause producers to shift current supply. If producers expect the price of their good to rise significantly in the near future, they may withhold current supply — storing output now to sell later at the higher price — causing the current supply curve to shift leftward. If they expect prices to fall, they may accelerate current production and selling, shifting the current supply curve rightward.

The diagrams below illustrate the two directions of supply shift:



The diagram shows three perfectly parallel supply curves — S0, S1 and S2 — each with the same slope, confirming that the shift is purely horizontal with no change in the responsiveness of producers to price. S1 is the original supply curve. S2 represents an increase in supply — a rightward shift — meaning that at every price level, including P1, producers are willing and able to supply a greater quantity Q2 rather than Q1. This could result from a fall in input costs, a technological improvement, a government subsidy, or an increase in the number of producers. S0 represents a decrease in supply — a leftward shift — meaning producers supply only Q0 at price P1. This could result from rising input costs, an indirect tax, adverse weather destroying productive capacity, or the exit of firms from the industry.

The key insight the diagram conveys is that at the unchanged price P1, the quantity supplied is entirely different across the three curves — Q0, Q1 and Q2 respectively. This is the defining characteristic of a supply shift as opposed to a movement along the supply curve. A movement along the curve occurs only when the price of the good itself changes, causing producers to move to a different point on the same curve. A shift, by contrast, occurs when any non-price determinant changes, moving the entire curve to a new position so that a different quantity is supplied at every price.

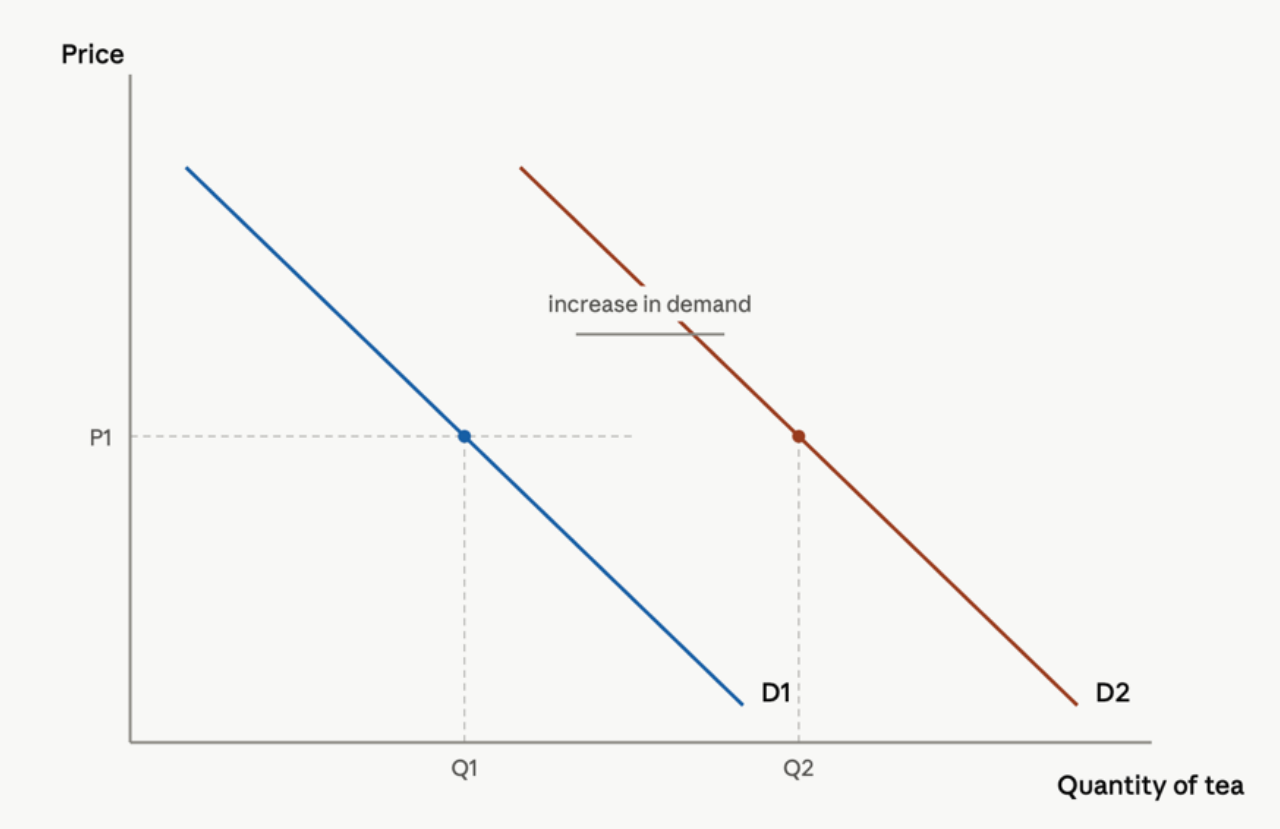
In conclusion, non-price determinants of supply — including input costs, technology, the number of producers, government taxes and subsidies, prices of related goods in production, and producer expectations — all cause shifts of the supply curve because they alter the profitability or productive capacity of firms independently of the market price. When these conditions improve, the curve shifts rightward to S2; when they deteriorate, it shifts leftward to S0. Understanding these shifts is essential for analysing how markets respond to changes in production conditions and why equilibrium prices and quantities change even when consumer demand remains constant.

Q4 Explain how a change in the price of a substitute good affects the demand for a product. [10]
Ans Demand refers to the quantity of a good or service that consumers are willing and able to buy at a given price over a given period of time, other things being equal. A substitute good is one that can be used in place of another to satisfy a similar want. Tea and coffee, Coca-Cola and Pepsi, or butter and margarine are classic examples. Because consumers see these goods as alternatives, the price of one has a direct influence on the demand for the other. Explaining this relationship properly requires distinguishing what happens to the good whose price changes from what happens to its substitute.

Consider two substitutes, tea and coffee. Suppose the price of coffee rises. For coffee itself, this causes a contraction (a movement up along its own demand curve) — quantity demanded of coffee falls because it has become more expensive. But the more important effect for this question concerns tea. As coffee becomes dearer, some consumers switch away from it and buy tea instead, since tea now looks relatively cheaper and offers similar satisfaction. At every possible price of tea, more tea is now demanded than before.

This is the key analytical point: a change in the price of a substitute does not cause a movement along the demand curve of the product in question — it causes the entire demand curve to shift. A rise in the price of coffee shifts the demand curve for tea to the right (an increase in demand),

while a fall in the price of coffee shifts the demand curve for tea to the left (a decrease in demand). The demand for the product changes even though the product's own price has not moved at all. The diagram below illustrates this rightward shift for tea following a rise in the price of coffee.



As the diagram shows, at the unchanged price P1 the quantity of tea demanded rises from Q1 to Q2 when the whole curve shifts from D1 to D2 — the increase in demand is caused entirely by the rise in the price of the substitute.

The relationship is therefore positive, or direct: the price of a substitute and the demand for a product move in the same direction. This is captured by a positive cross elasticity of demand between the two goods. The stronger the substitute relationship, the larger the shift; close substitutes such as two competing soft-drink brands produce a big response, whereas weak substitutes produce only a small one.

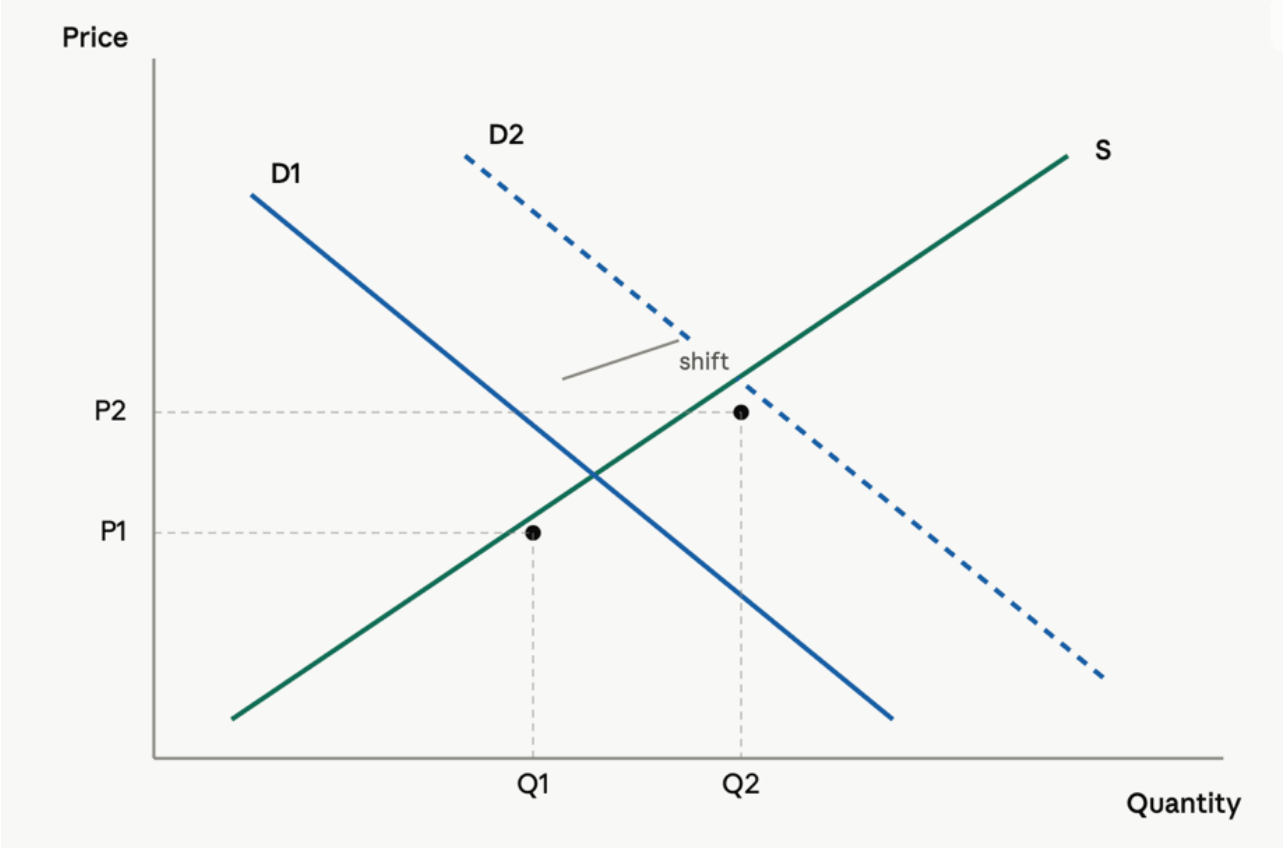
A few qualifying points strengthen the analysis. The effect depends on how close the substitutes really are — goods that are only loosely interchangeable will show a muted response. It also assumes the ceteris paribus condition holds, meaning income, tastes, and the prices of other goods remain unchanged; in reality several determinants may move at once. Finally, the size of the shift depends on consumers' willingness and ability to switch, which can be limited by habit, brand loyalty, or contracts.

In summary, a change in the price of a substitute affects a product's demand by shifting its whole demand curve rather than moving along it. A rise in the substitute's price raises demand for the product, and a fall in the substitute's price lowers it — a positive relationship reflected in a positive cross elasticity of demand.

Q5 Using a demand and supply diagram, discuss the impact of an increase in consumer income on the equilibrium price and quantity of a normal good. [15]

A normal good is one for which demand rises as consumer income increases, such as restaurant meals, cars, or air travel. Demand refers to the quantity of a good consumers are willing and able to buy at each price over a period of time, while supply is the quantity producers are willing to offer for sale at each price. Market equilibrium occurs where quantity demanded equals quantity supplied, giving a single equilibrium price and quantity with no shortage or surplus. This answer examines what happens to that equilibrium when income rises.

When consumer income increases, households have greater purchasing power. For a normal good, this raises the quantity consumers wish to buy at every price, so the entire demand curve shifts to the right, from D1 to D2. Income is a non-price determinant of demand, so it shifts the whole curve rather than causing a movement along it. The supply curve does not move, because none of its own determinants — costs of production, technology, or the number of firms — has changed. This is shown in the diagram below.



Initially the market is in equilibrium where D1 intersects S, at price P1 and quantity Q1. After the rightward shift to D2, at the old price P1 the quantity demanded now exceeds the quantity supplied, creating a shortage. This shortage puts upward pressure on price. As price rises, producers increase the quantity supplied (a movement up along S) while the higher price chokes off some of the extra demand (a movement along D2). The market settles at a new equilibrium

where D2 crosses S, at a higher price P2 and a higher quantity Q2. The clear result is that an increase in income raises both the equilibrium price and the equilibrium quantity of a normal good, because it is the demand curve, not the supply curve, that has shifted.

A real-world example is the market for air travel in India. As household incomes have risen over the past decade, air travel — a normal good — has seen demand climb strongly, with more people choosing to fly rather than take the train. This rightward shift in demand has supported both higher passenger volumes and firmer fares on popular routes, particularly during peak periods, illustrating the theoretical prediction of a higher price and quantity.

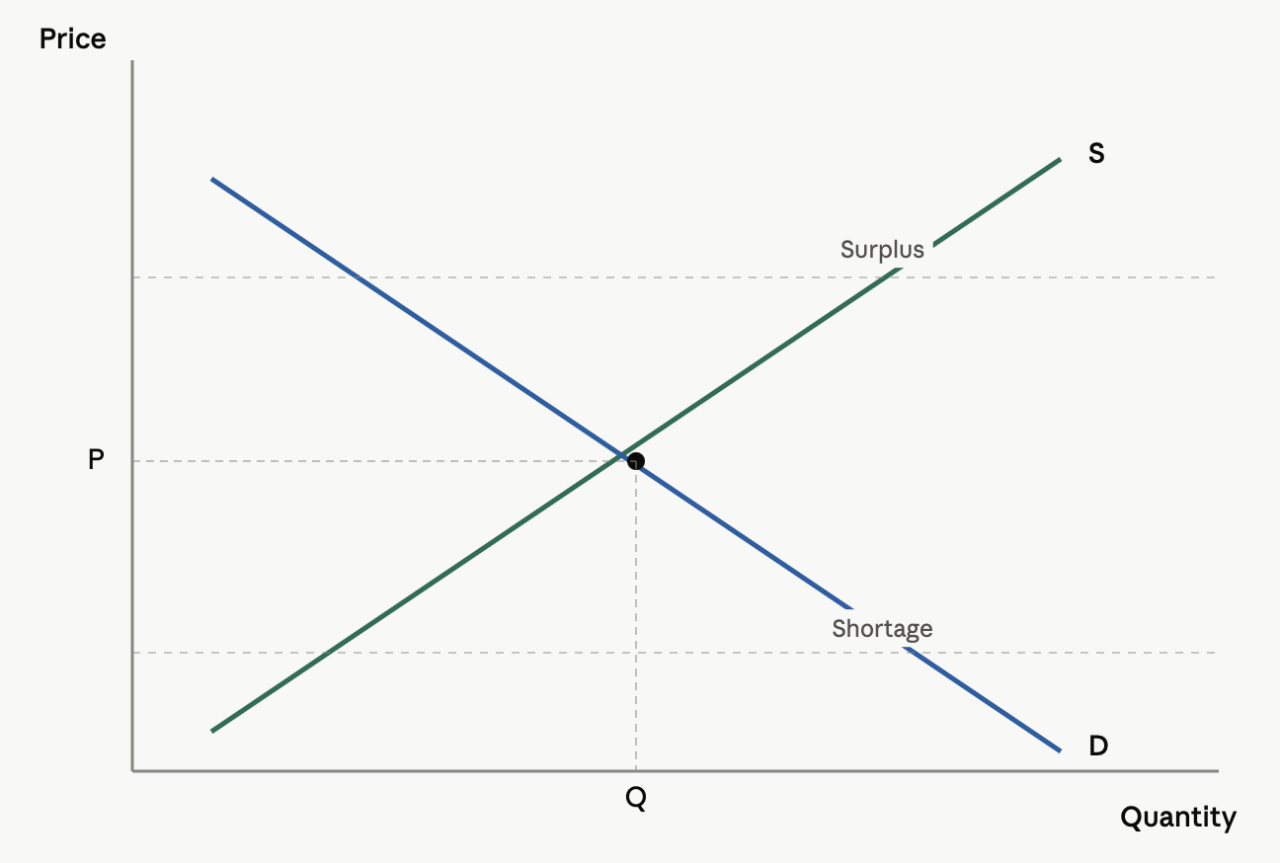
However, the analysis requires evaluation. The size of the changes in price and quantity depends on the price elasticity of supply. If supply is elastic, because producers can easily expand output, most of the adjustment appears as a large rise in quantity with only a small rise in price. If supply is inelastic in the short run, as with airport slots or prime retail locations, the same demand shift produces a larger price rise and a smaller quantity increase. The magnitude of the demand shift itself depends on the income elasticity of demand: income-elastic luxuries shift far more than income-inelastic necessities for a given income change.

The conclusion also rests on the ceteris paribus assumption. In reality, rising incomes may coincide with changing tastes, new firms entering, or altered input costs, any of which could shift supply as well and complicate the outcome. Furthermore, a good classed as normal at lower income levels may become inferior at higher ones, in which case demand would fall rather than rise. The direction of the effect therefore depends on where consumers sit on the income scale.

In summary, for a normal good an increase in consumer income shifts demand to the right and raises both equilibrium price and quantity. The relative size of these two changes is determined mainly by the price elasticity of supply and the income elasticity of demand, and the result holds only under the assumption that other influences on the market remain unchanged.

Q6 Explain how the law of demand and the law of supply interact to determine the market equilibrium. [10]
Ans The law of demand states that, other things being equal, as the price of a good falls the quantity demanded rises, and as price rises the quantity demanded falls. This gives the demand curve its downward slope, reflecting how consumers buy more when goods become cheaper. The law of supply states the opposite relationship: as price rises the quantity supplied rises, and as price falls the quantity supplied falls, because higher prices make production more profitable and encourage firms to offer more for sale. This gives the supply curve its upward slope. The two laws pull in opposite directions with respect to price, and it is precisely this opposition that allows a single equilibrium to emerge.

Market equilibrium is the price at which the quantity that consumers wish to buy exactly equals the quantity that producers wish to sell. Graphically it is the point where the downward-sloping demand curve intersects the upward-sloping supply curve. At this price there is neither a shortage nor a surplus, so there is no pressure for the price to change; the market is said to clear.



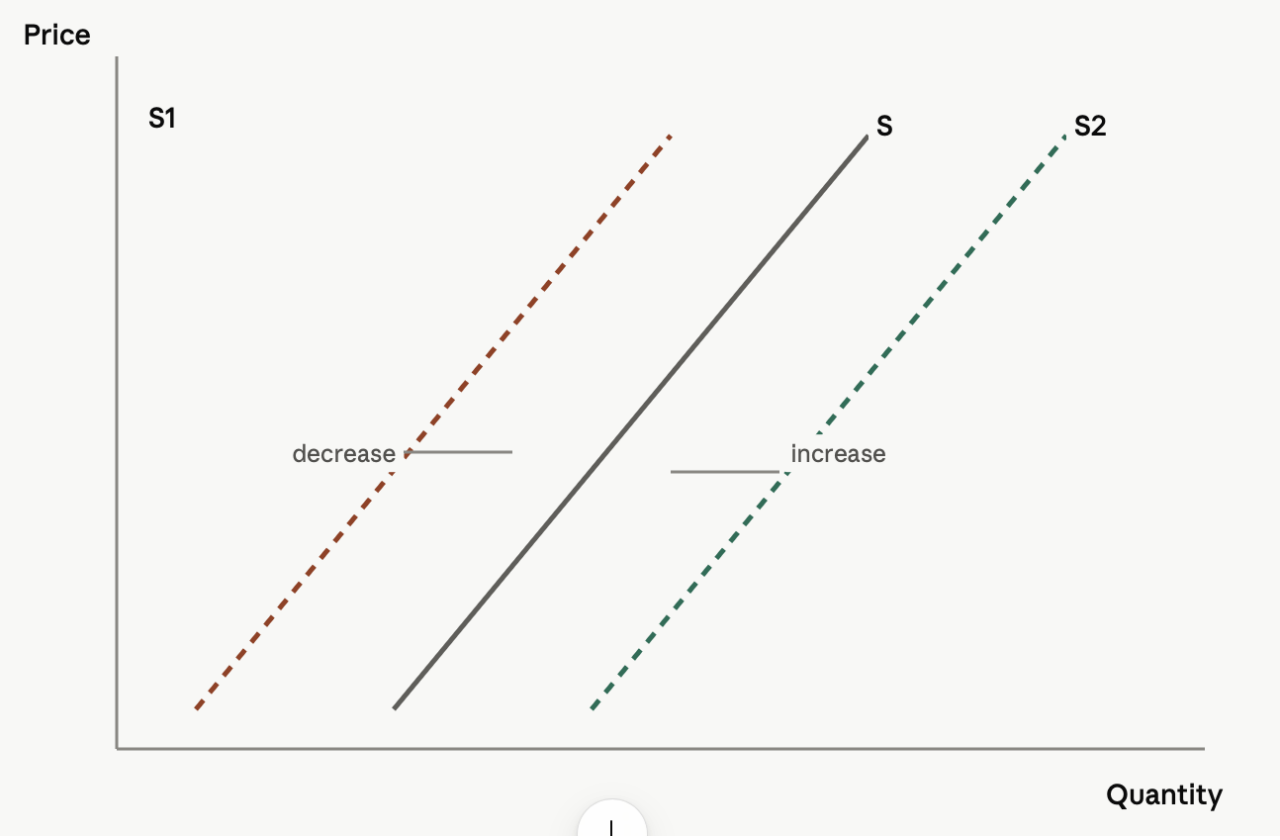
The way the two laws interact to reach this point is best understood by considering what happens when the price is not at equilibrium. Suppose the price is set above equilibrium. At this higher price the law of supply means producers offer a large quantity, while the law of demand means consumers want only a small quantity. The result is a surplus, or excess supply. To clear their unsold stock, sellers cut their prices. As the price falls, the quantity demanded rises (moving along the demand curve) and the quantity supplied falls (moving along the supply curve), and this continues until the surplus is eliminated at the equilibrium price.

The opposite happens when the price is set below equilibrium. At this lower price the law of demand means consumers want a large quantity, while the law of supply means producers offer only a small quantity, creating a shortage, or excess demand. Competition among buyers for the limited goods bids the price up. As the price rises, the quantity demanded falls and the quantity supplied rises, until the shortage disappears and equilibrium is restored.

In this way the two laws act as automatic, self-correcting forces. Whenever the price is too high, the surplus it creates drives the price down; whenever the price is too low, the shortage it creates drives the price up. The market naturally gravitates towards the single price where the quantity demanded equals the quantity supplied. This is what economists call the price mechanism, sometimes described as the "invisible hand," through which the independent decisions of buyers and sellers are coordinated. In summary, the law of demand and the law of supply interact through the pressure of surpluses and shortages to guide the market towards equilibrium, the only price at which the plans of consumers and producers are exactly compatible.

Q7 Explain how non-price determinants of supply can lead to a shift of the supply curve. [10]
Ans The law of supply describes how the quantity supplied of a good responds to a change in its own price, shown as a movement along a fixed supply curve. However, price is not the only thing that influences how much producers are willing to offer for sale. The other factors are known as the non-price determinants of supply, or the conditions of supply. When any of these change, the whole supply curve shifts to a new position, because producers are now willing to supply a different quantity at every possible price. This is fundamentally different from a movement along the curve: a change in the good's own price moves us along the curve, whereas a change in a non-price determinant shifts the entire curve.

A rightward shift represents an increase in supply, meaning more is supplied at each price; a leftward shift represents a decrease in supply, meaning less is supplied at each price. The diagram below illustrates both cases.



The main non-price determinants that cause these shifts can be explained as follows.

The first is the **cost of production**. Supply depends heavily on the prices of inputs such as raw materials, wages, energy, and rent. If input costs fall — say cheaper raw materials — production becomes more profitable at any given price, so firms supply more and the curve shifts right. If input costs rise, profitability falls, firms supply less, and the curve shifts left.

The second is **technology**. An improvement in technology raises productivity, allowing firms to produce more output from the same inputs at lower unit cost. This increases supply and shifts the curve to the right. Conversely, a loss of technology or breakdown of equipment would reduce supply.

A third factor is **taxes and subsidies**. An indirect tax on production acts like an additional cost, reducing profitability and shifting supply to the left. A subsidy has the opposite effect: it lowers the effective cost of production, encouraging firms to supply more and shifting the curve to the right.

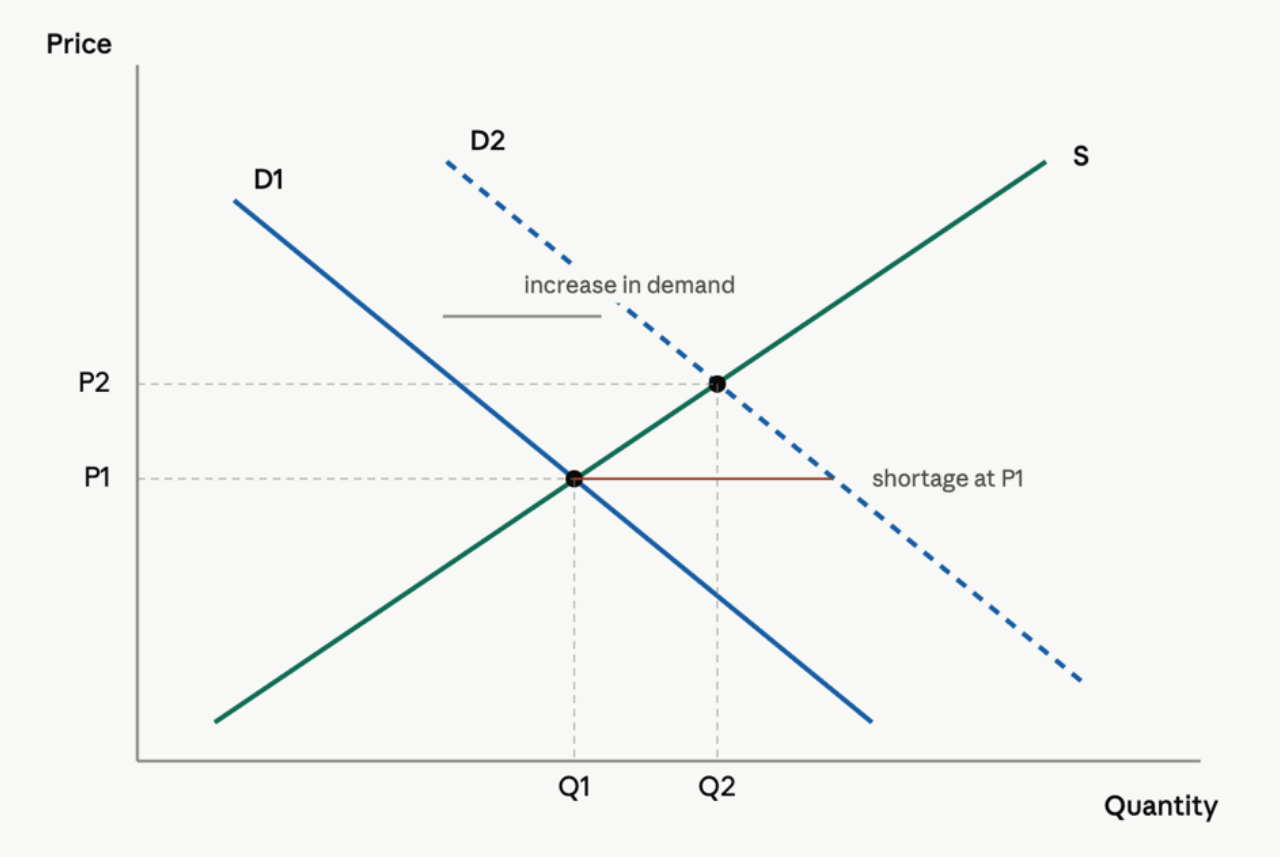
A fourth is the **number of firms** in the market. If new firms enter an industry, total market supply increases and the curve shifts right. If firms leave, perhaps because they are making losses, market supply falls and the curve shifts left.

Other determinants include the prices of related goods in production. If a firm can use its resources to produce an alternative good that becomes more profitable, it may switch production, reducing the supply of the original good. Supply is also affected by expectations of future prices, by weather and natural conditions (especially important in agriculture, where a good harvest shifts supply right and a drought shifts it left), and by unexpected events or shocks such as strikes or disruptions to supply chains.

In each of these cases the key point is the same: because the change alters how much producers are willing to offer at every price rather than responding to the good's own price, the effect is a shift of the entire supply curve rather than a movement along it. In summary, non-price determinants such as production costs, technology, taxes and subsidies, the number of firms, and natural conditions change the underlying willingness to supply, shifting the supply curve rightward when supply increases and leftward when supply decreases.

Q8 Explain how the price mechanism reallocates resources when there is an increase in demand for a good or a service.
Ans The price mechanism is the way in which the forces of demand and supply interact through prices to allocate scarce resources in a free market. Prices carry out three key functions in this process: **they act as signals that convey information about changing market conditions, as incentives that reward producers for responding, and as a rationing device that distributes goods to those willing and able to pay**. An increase in demand for a good sets all three of these functions in motion, and the outcome is that resources are reallocated towards producing more of that good. The following explains the sequence of events.

Suppose there is an increase in demand for a good, perhaps because it has become more fashionable, incomes have risen, or the price of a substitute has increased. **At the original price, the quantity now demanded exceeds the quantity supplied, so a shortage develops. This shortage is the trigger that starts the reallocation process.**



Under your price mechanism comes three processes :

Signal – Due to an increase in demand, there will be shortage. This signals that the good has become scarce.
Incentive – Due to shortage, the price will go up. This incentivises the sellers or producers to produce more as it will be profitable for them to supply more.
Rationing – Due to an increase in price, some consumers will ration it means they won't be able to buy the good at a higher price.

The signalling and rationing function comes first. **The shortage causes the price to rise, as consumers compete with one another for the limited supply and producers realise they can charge more. In the diagram, the demand curve shifts from D1 to D2 and the price is bid up from P1 to P2. This rising price rations the good in the short run, so that it goes to those consumers who value it most and are willing to pay the higher price. At the same time, the higher price acts as a signal to producers that this good has become more desirable to consumers.**

The incentive function comes next. **The higher price of P2 raises the profitability of producing this good, which gives existing firms an incentive to expand output and encourages new**

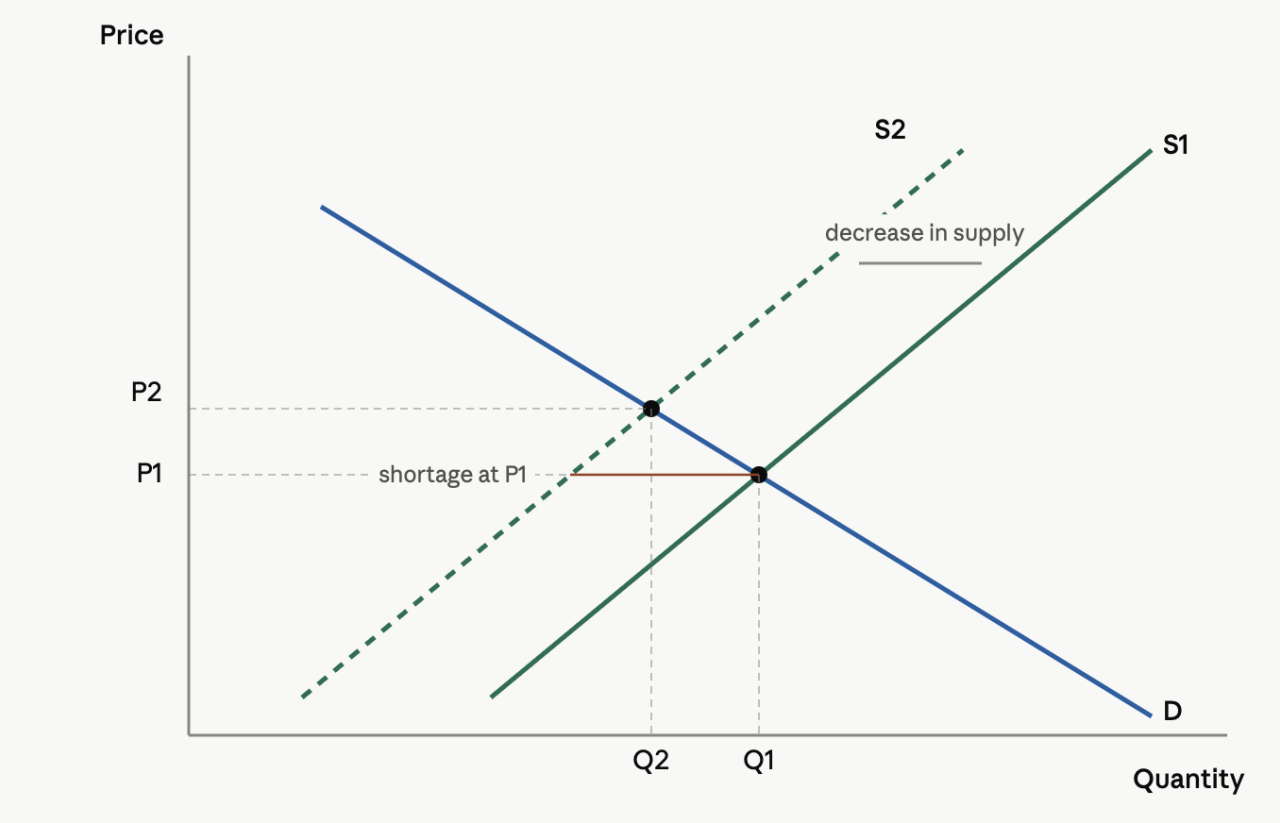
firms to enter the market. Producers respond by increasing the quantity supplied, shown as a movement up along the supply curve from Q1 to Q2. To achieve this extra output, firms must acquire more factors of production — labour, raw materials, machinery, and land — and so resources are drawn into this industry.

This is the heart of the reallocation. Because resources are scarce, the factors of production attracted into this expanding industry must come from elsewhere in the economy. Other industries, where demand has not risen and profitability is comparatively lower, will find it harder to bid for those resources and will contract. In this way the price mechanism shifts land, labour, and capital out of goods that consumers want relatively less and into the good that consumers now want more. The economy ends up producing more of what consumers value, in line with the change in their preferences.

The market settles at a new equilibrium at price P2 and quantity Q2, where the quantity demanded once again equals the quantity supplied and the shortage has been eliminated. More of society's resources are now devoted to producing this good than before.

In summary, an increase in demand creates a shortage that pushes up the price; the higher price signals the change to producers, rations the good among consumers, and raises profits, which incentivises firms to expand output and draw factors of production away from other uses. Through this chain of signals, incentives, and rationing, the price mechanism automatically reallocates scarce resources towards the good that consumers now demand more of, without any central direction.

Q9 Explain how the price mechanism reallocates resources when there is a decrease in supply of a good. (10)
Ans The price mechanism allocates scarce resources through the signalling, incentive, and rationing functions of prices. When the supply of a good decreases, these functions work in reverse compared with a demand increase: resources are drawn away from the good rather than towards it. **A decrease in supply means producers are willing to offer less at every price, shown as a leftward shift of the supply curve. This can happen for reasons such as a rise in production costs, a bad harvest, an increase in an indirect tax, or firms leaving the industry.**



As the diagram shows, the supply curve shifts left from S1 to S2 while demand D stays fixed. At the original price P1, the quantity now supplied is less than the quantity demanded, so a shortage develops. **This shortage is what sets the reallocation in motion.**

The signalling and rationing function acts first. **The shortage causes consumers to compete for the reduced quantity available, bidding the price up from P1 to P2. This higher price rations the smaller supply, so the good goes to those consumers who value it most and are still willing to pay. The rising price also signals to the market that the good has become relatively scarcer.**

The incentive and reallocation function follows. **Although the higher price would normally attract producers, the point is that supplying this good has become more costly or difficult — that is why supply fell in the first place. Firms move up along the new supply curve S2 as price rises, but the market settles at a lower equilibrium quantity Q2 than before. Because less of the good is now produced, fewer factors of production** are needed in this industry, so land, labour, and capital are released from it**. These resources are then free to move into other industries where they can earn a better return, and the economy ends up devoting fewer resources to this good.**

The new equilibrium is at price P2 and quantity Q2, where the quantity demanded once again equals the quantity supplied and the shortage has been eliminated. Society now produces less of this good and has reallocated some of its scarce resources elsewhere.

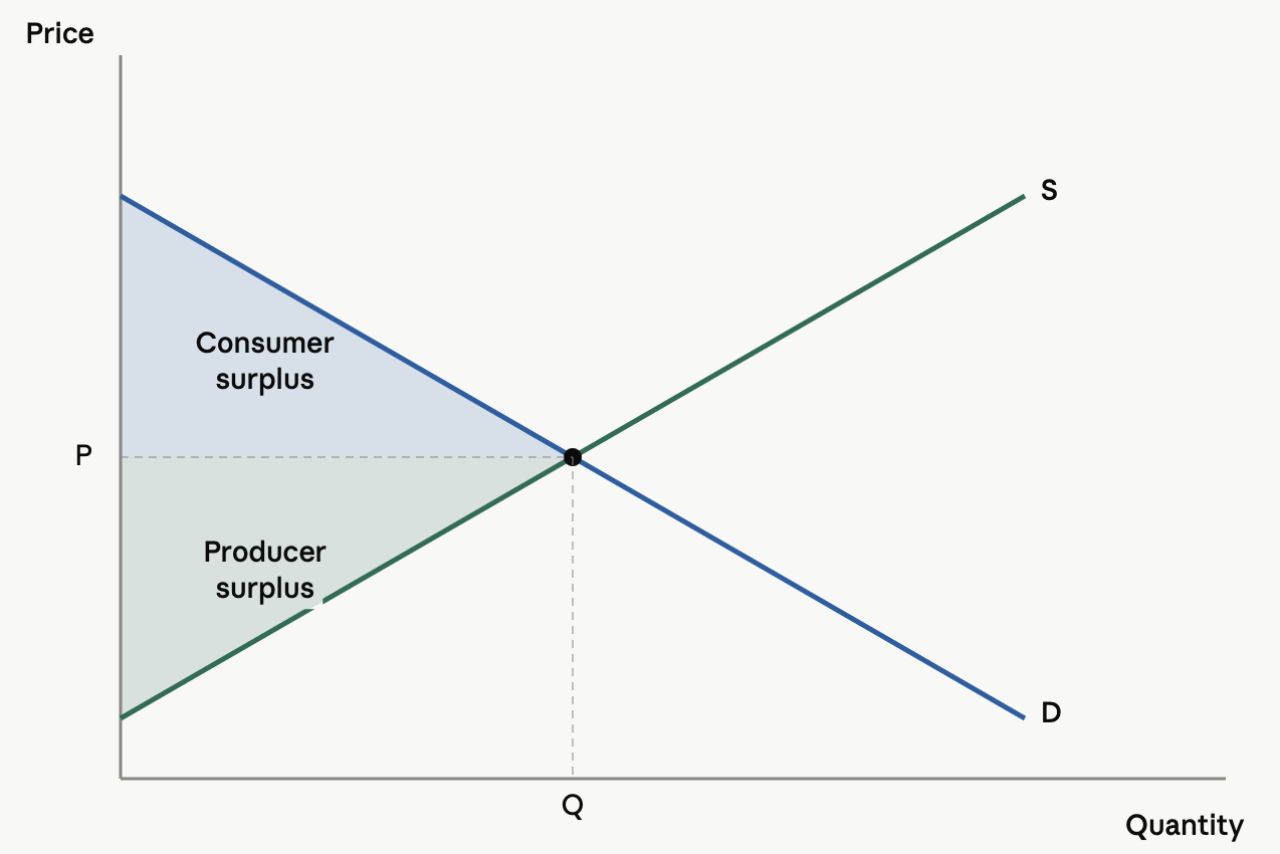
In summary, a decrease in supply creates a shortage that pushes the price up; the higher price rations the reduced quantity among consumers and signals greater scarcity, while the fall in output means fewer resources are required in the industry. Through this process the price mechanism

moves scarce factors of production out of the good whose supply has fallen and into alternative uses, without any central direction.

Q10 Explain the concept of consumer and producer surplus. (10) Ans Consumer surplus and producer surplus are measures of the economic welfare, or benefit, that buyers and sellers gain from participating in a market. They are best understood together, using a demand and supply diagram, because each is a triangular area measured from the equilibrium price.

Consumer surplus is the difference between the maximum price a consumer is willing and able to pay for a good and the price they actually pay. The demand curve shows the highest price each consumer is prepared to pay, reflecting the value they place on the good. Because there is a single market price, consumers who would have been willing to pay more than that price enjoy a "bonus" — they get the good for less than it was worth to them. Adding up this gain across all consumers gives total consumer surplus, shown as the area below the demand curve and above the market price.

Producer surplus is the mirror image: it is the difference between the price a producer actually receives and the minimum price they would have been willing to accept to supply the good. The supply curve shows this minimum acceptable price, which reflects the cost of production. Producers who would have been willing to sell for less than the market price gain a surplus, because they receive more than their lowest acceptable price. Summed across all producers, this is total producer surplus, shown as the area above the supply curve and below the market price.



As the diagram shows, at the equilibrium price P and quantity Q the two surpluses sit one above the other. The upper triangle, between the demand curve and the price line, is consumer surplus; the lower triangle, between the price line and the supply curve, is producer surplus. The demand curve represents consumers' willingness to pay and the supply curve represents producers' willingness to accept, so the vertical gap between each curve and the price P measures the benefit gained on every unit up to Q.

The sum of consumer surplus and producer surplus is known as total welfare, or community surplus, and it is maximised at the free-market equilibrium. This is why the equilibrium point is regarded as allocatively efficient: the combined benefit to buyers and sellers is as large as possible.

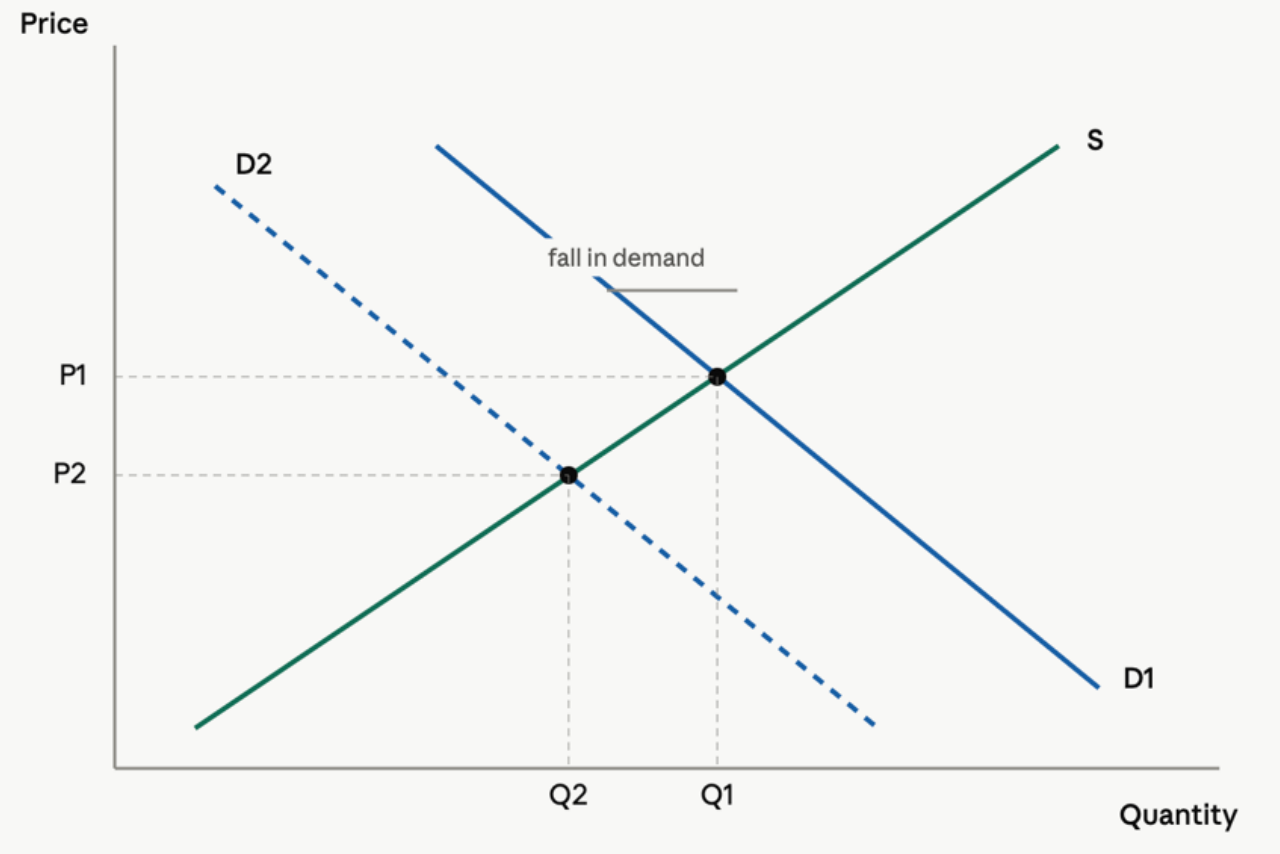
It is also useful to note how the two surpluses change. If the market price falls, consumer surplus rises (consumers keep more of their willingness to pay) while producer surplus falls, and vice versa. Shifts in demand or supply also change the size of the triangles: for example, an increase in demand tends to raise producer surplus, while an increase in supply that lowers price tends to raise consumer surplus.

In summary, consumer surplus is the benefit consumers gain from paying less than they were willing to, and producer surplus is the benefit producers gain from receiving more than their minimum acceptable price. Represented as the areas between the price and the demand and supply curves respectively, together they measure the total welfare generated by a market and are greatest at the competitive equilibrium.

Q13 Explain how a change in the price of a complementary good affects the demand for a product. [10]
Ans A complementary good is one that is consumed together with another good, so that the two are used jointly to satisfy a want. Classic examples are cars and petrol, printers and ink cartridges, mobile phones and data plans, or tennis rackets and tennis balls. Because these goods are demanded together, a change in the price of one has a direct effect on the demand for the other. Demand here refers to the quantity of a good consumers are willing and able to buy at each price over a period of time.

The key point is that the price of a complement is a non-price determinant of demand for the product. A change in it therefore shifts the whole demand curve for the product, rather than causing a movement along it. The relationship is negative, or inverse: the price of a complement and the demand for the product move in opposite directions.

Consider cars and petrol. Suppose the price of petrol rises sharply. Petrol and cars are used together, so the rise in the running cost of motoring makes owning and using a car less attractive. As a result, at every possible price of cars, consumers now wish to buy fewer cars than before. The demand curve for cars shifts to the left, from D1 to D2, representing a fall in demand. Importantly, the price of cars themselves has not changed — it is the price of the complement that has triggered the shift.



The diagram shows this fall in demand. The demand curve for the product shifts left from D1 to D2. At the original equilibrium the price and quantity were P1 and Q1; after the leftward shift they fall to P2 and Q2. The whole curve has moved because the change was caused by the price of a related good, not by the product's own price.

The opposite case works in the same way. If the price of the complement falls — say petrol becomes much cheaper — motoring becomes cheaper overall, cars become more attractive to own, and the demand for cars rises. The demand curve for cars would then shift to the right, raising both the equilibrium price and quantity.

This inverse relationship is captured by the cross elasticity of demand between two complements, which is negative. The stronger the complementary relationship, the larger the shift in demand for a given change in the complement's price. Close complements such as game consoles and their games produce a large effect, whereas weak complements produce only a small one.

A few qualifying points strengthen the analysis. The size of the effect depends on how strongly the goods are complements and on how large the price change is. The analysis also assumes ceteris paribus — that other determinants of demand such as income and tastes stay constant. In reality several determinants may change at once, which can offset or reinforce the effect.

In summary, a change in the price of a complementary good shifts the entire demand curve for the product in the opposite direction: a rise in the complement's price reduces demand for the product and shifts the curve left, while a fall in the complement's price increases demand and shifts the curve right. This negative relationship is reflected in a negative cross elasticity of demand.

Q14 Explain how a decrease in income might affect the demand for normal goods and the demand for inferior goods. (15M)
Ans Income is one of the key non-price (conditional) determinants of demand. A change in income, holding price constant, causes a *shift* of the entire demand curve rather than a movement along it. The direction of that shift depends on whether the good is normal or inferior, which is measured by its **income elasticity of demand (YED)** — the responsiveness of quantity demanded to a change in income, calculated as %ΔQd ÷ %ΔY.

A **normal good** is one for which demand and income move in the same direction, so YED is positive. An **inferior good** is one for which demand and income move in opposite directions, so YED is negative. A fall in income therefore has opposite effects on the two.

Effect on normal goods

For a normal good, a fall in income reduces consumers' purchasing power, so at every price level they are willing and able to buy less. Demand contracts and the demand curve shifts to the **left**, from D■ to D■, as shown below.

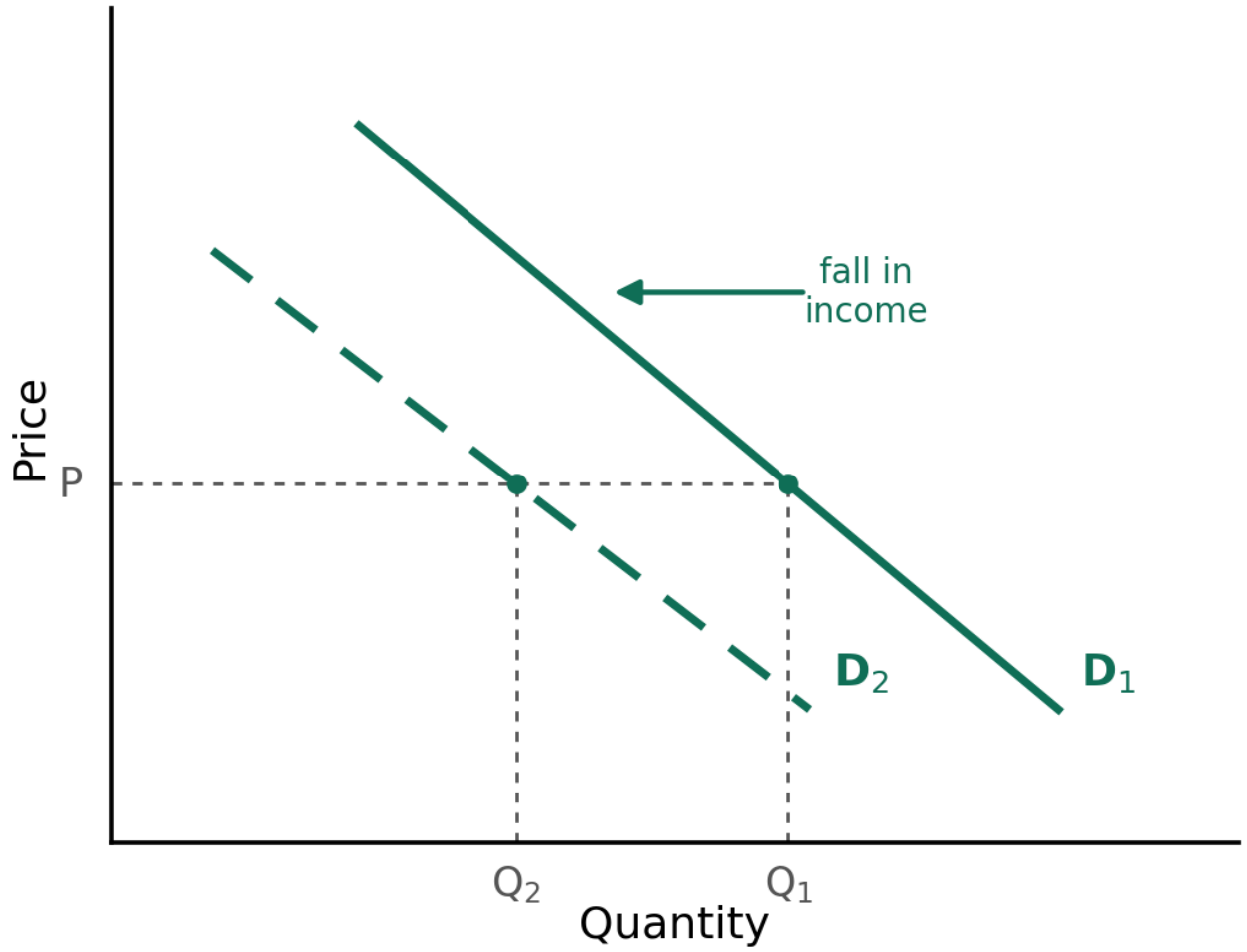


Figure 1: A fall in income shifts the demand for a normal good leftward (D■ → D■)

In Figure 1, the original demand curve is D■. Because income has fallen while price is unchanged, the whole curve shifts inward to D■, which lies to the left of and parallel to D■. This is a decrease in demand, not a contraction along the curve. Taking any given price P, the quantity demanded falls from Q■ (read off D■) to Q■ (read off D■): at the *same* price, consumers now buy less of the normal good.

The size of this contraction depends on the value of YED. **Luxuries** have a YED greater than 1, so demand falls proportionately more than income — a fall in income might sharply cut demand for foreign holidays, restaurant meals or new cars. **Necessities** have a YED between 0 and 1 (e.g. bread, electricity), so demand falls only slightly, because consumers protect essential spending.

Effect on inferior goods

For an inferior good, the same fall in income raises demand. As consumers become poorer they substitute away from more expensive alternatives and switch *towards* cheaper options they previously avoided, so at every price they buy more. The demand curve shifts to the **right**, from D■ to D■.

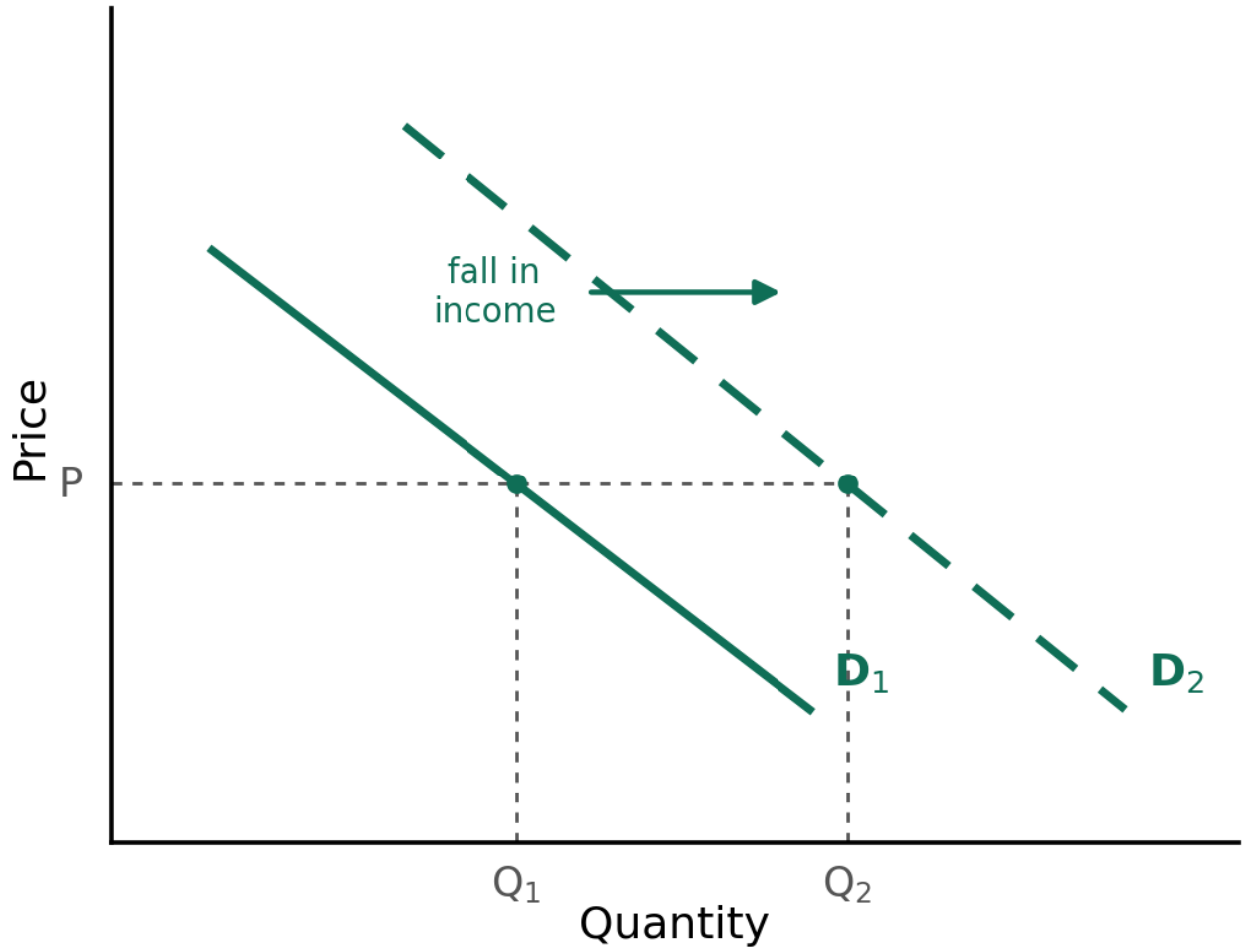


Figure 2: A fall in income shifts the demand for an inferior good rightward (D■ → D■)

Figure 2 shows the opposite movement. Starting again from D■, the fall in income shifts the demand curve outward to D■, to the right of and parallel to D■. This is an increase in demand: at the same price P, the quantity demanded rises from Q■ to Q■, so consumers buy *more* of the inferior good as they trade down towards cheaper substitutes. Comparing the two figures makes the contrast clear — at an identical price, the same fall in income moves quantity demanded in *opposite* directions depending on the sign of the good's income elasticity.

Typical examples are supermarket own-brand products, instant noodles, second-hand clothing and public bus travel — bought *more* as income falls because they replace the pricier substitutes households can no longer afford. This explains why discount retailers such as Aldi, Lidl or Poundland often gain sales during recessions.

Evaluation

However, the effect is not always so clear-cut. First, whether a good is normal or inferior depends on the consumer and their income level: a car may be a luxury for a low earner but a necessity for a high earner, and bus travel may be inferior for the wealthy yet normal for those on very low incomes. Classification is therefore not fixed.

Second, the *magnitude* of the shift matters more than its direction — for essential normal goods with inelastic YED, the leftward shift may be almost negligible. Third, the analysis assumes ceteris paribus; in reality a fall in income rarely occurs in isolation. During a recession, prices, expectations and confidence also change, which may reinforce or offset the income effect. Finally, a good can switch from normal to inferior beyond a certain income threshold, so the same good may behave differently for the same consumer over time.

Conclusion

In short, a fall in income shifts the demand curve *leftward* for normal goods and *rightward* for inferior goods — the two being distinguished by the sign of their income elasticity of demand. The strength of each effect ultimately depends on the value of YED, the consumer's circumstances, and conditions in the wider market.

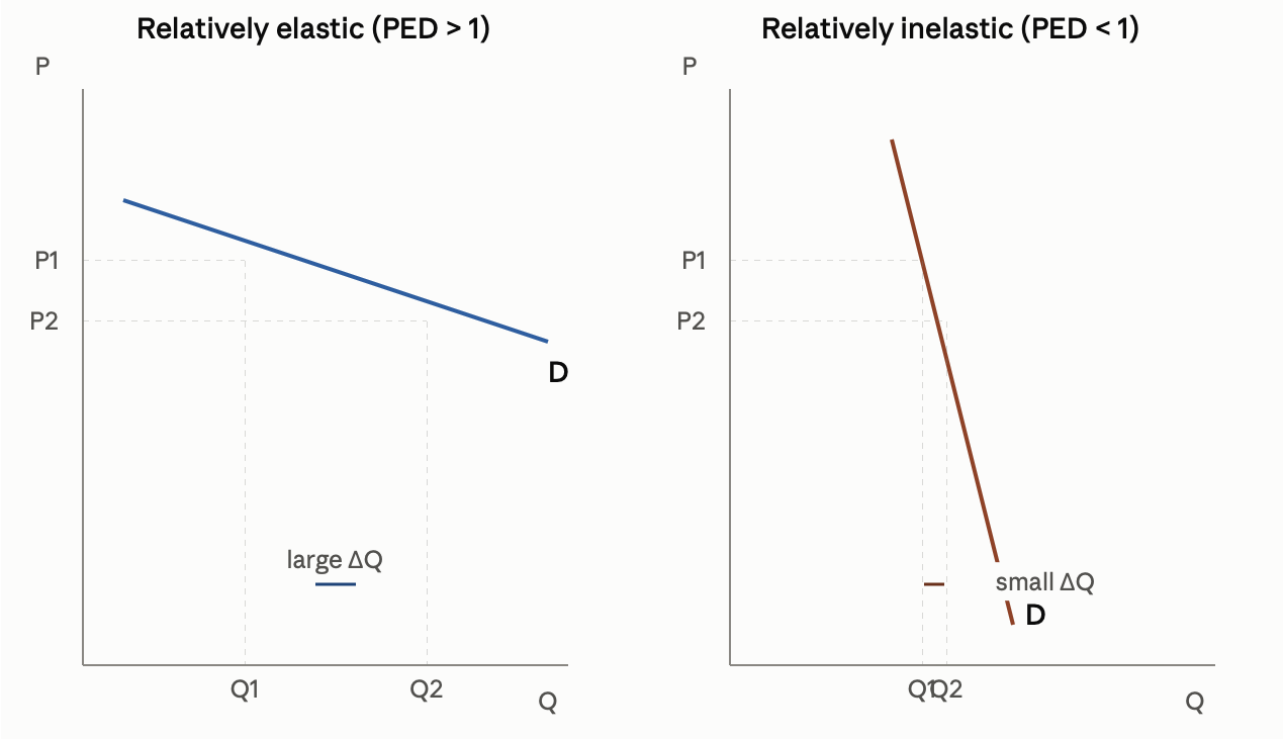
Q15 Explain two determinants of price elasticity of demand. (10) (2023 Nov HL)
Ans Price elasticity of demand (PED) measures the responsiveness of the quantity demanded of a good to a change in its own price. It is calculated as the percentage change in quantity demanded divided by the percentage change in price, and because demand curves normally slope downward the value is negative, though it is usually referred to by its absolute size. Demand is described as price elastic when PED is greater than 1 (quantity demanded changes proportionally more than price) and price inelastic when PED is between 0 and 1 (quantity demanded changes proportionally less than price). The value of PED is not the same for every good; it depends on a number of determinants, two of which — the availability of close substitutes and the proportion of income spent on the good — are explained below.

The first and generally most important determinant is the **number and closeness of substitutes**. A substitute is a good that can be used in place of another to satisfy a similar want. When a good has many close substitutes, its demand tends to be price elastic, because if its price rises consumers can easily switch to an alternative, causing quantity demanded to fall by a large proportion. For example, a particular brand of soft drink has many close substitutes, so a rise in its price leads many consumers to buy a rival brand instead, and demand is elastic. By contrast, a good with few or no close substitutes tends to have price inelastic demand, because consumers who want it have little choice but to keep buying it even when its price rises. Petrol is a common example: in the short run there is no convenient substitute for most drivers, so a rise in price causes only a small proportional fall in quantity demanded. The closeness of substitutes also depends on how the good is defined — demand for "a Coca-Cola" is far more elastic than demand for "soft drinks in general," which in turn is more elastic than demand for "drinks," because the more broadly a good is defined the fewer substitutes lie outside the category.

The second determinant is the **proportion of income spent on the good**. This refers to how large a share of a consumer's total income is taken up by spending on that good. Goods that account for only a very small proportion of income tend to have price inelastic demand, because even a large percentage change in their price has a negligible effect on the consumer's overall budget, so quantity demanded barely changes. Salt and matches are classic examples: their prices could rise substantially yet consumers would hardly alter the quantity they buy, because the

amount spent is trivial relative to income. Conversely, goods that take up a large proportion of income tend to have more price elastic demand, because a change in their price has a noticeable impact on how much a consumer can afford. A rise in the price of an expensive item such as a car or an overseas holiday absorbs a significant share of income, so consumers are more likely to respond by cutting back, delaying the purchase, or seeking a cheaper alternative, making quantity demanded relatively responsive to price.

These two determinants can be illustrated using demand curves of differing steepness, as shown in the diagram below: a relatively flat (elastic) demand curve represents a good with many substitutes or one that takes a large share of income, where a given price change produces a large change in quantity demanded, while a relatively steep (inelastic) demand curve represents a good with few substitutes or a small share of income, where the same price change produces only a small change in quantity demanded.

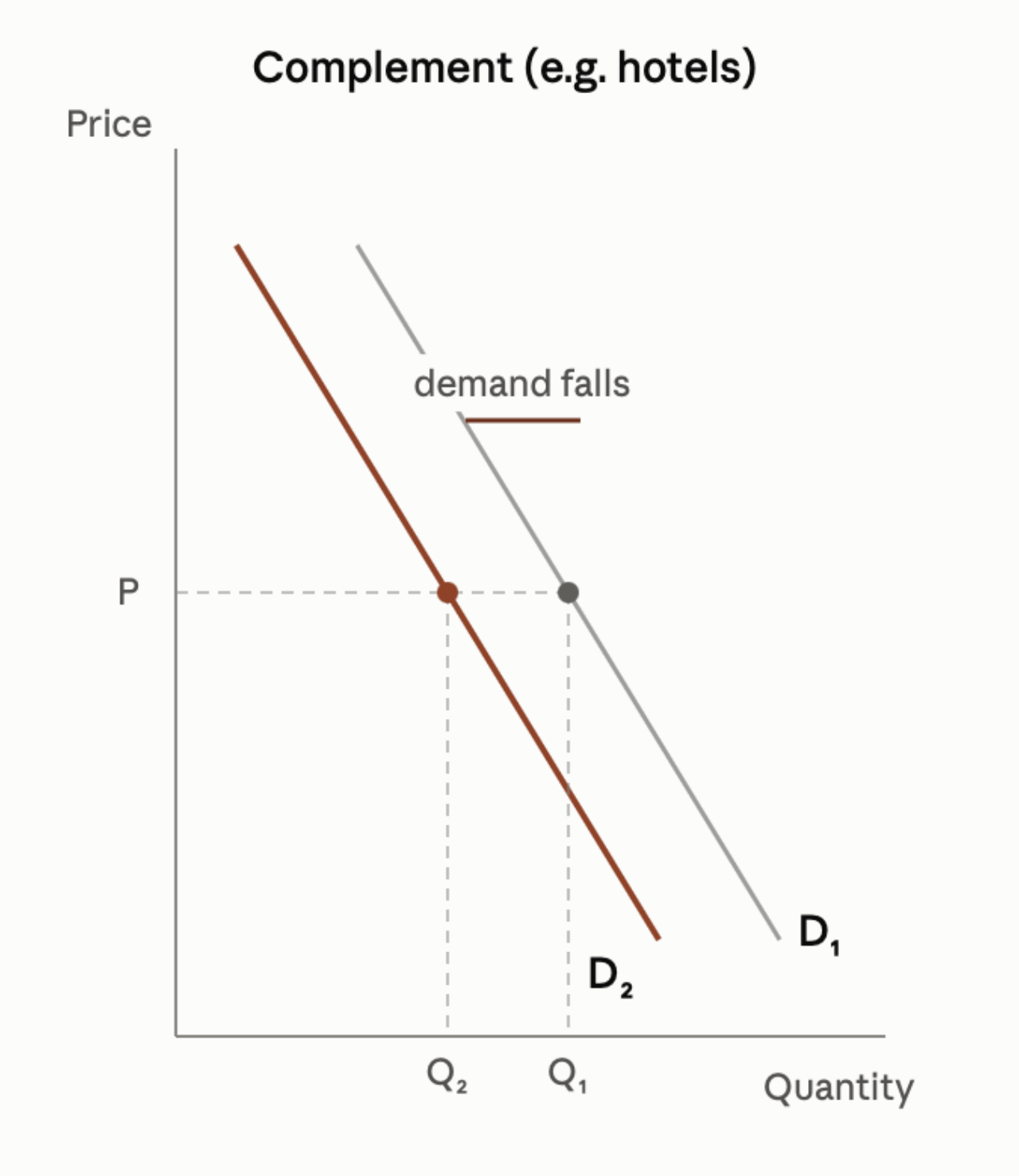


the same fall in price from P■ to P■ produces a **large** rise in quantity demanded (Q■ to Q■) along the flat, elastic curve on the left — the case of a good with many substitutes or a large share of income — but only a **small** rise along the steep, inelastic curve on the right, the case of few substitutes or a small share of income.

In conclusion, price elasticity of demand is determined by several factors, of which the number and closeness of substitutes and the proportion of income spent on the good are two of the most significant. Demand tends to be more elastic when a good has many close substitutes and takes up a large proportion of income, and more inelastic when substitutes are few and the good absorbs only a small share of income. Because most real-world goods sit somewhere between these extremes, and because these determinants can pull in different directions, the actual PED of any good reflects the combined influence of all its determinants rather than any single one.

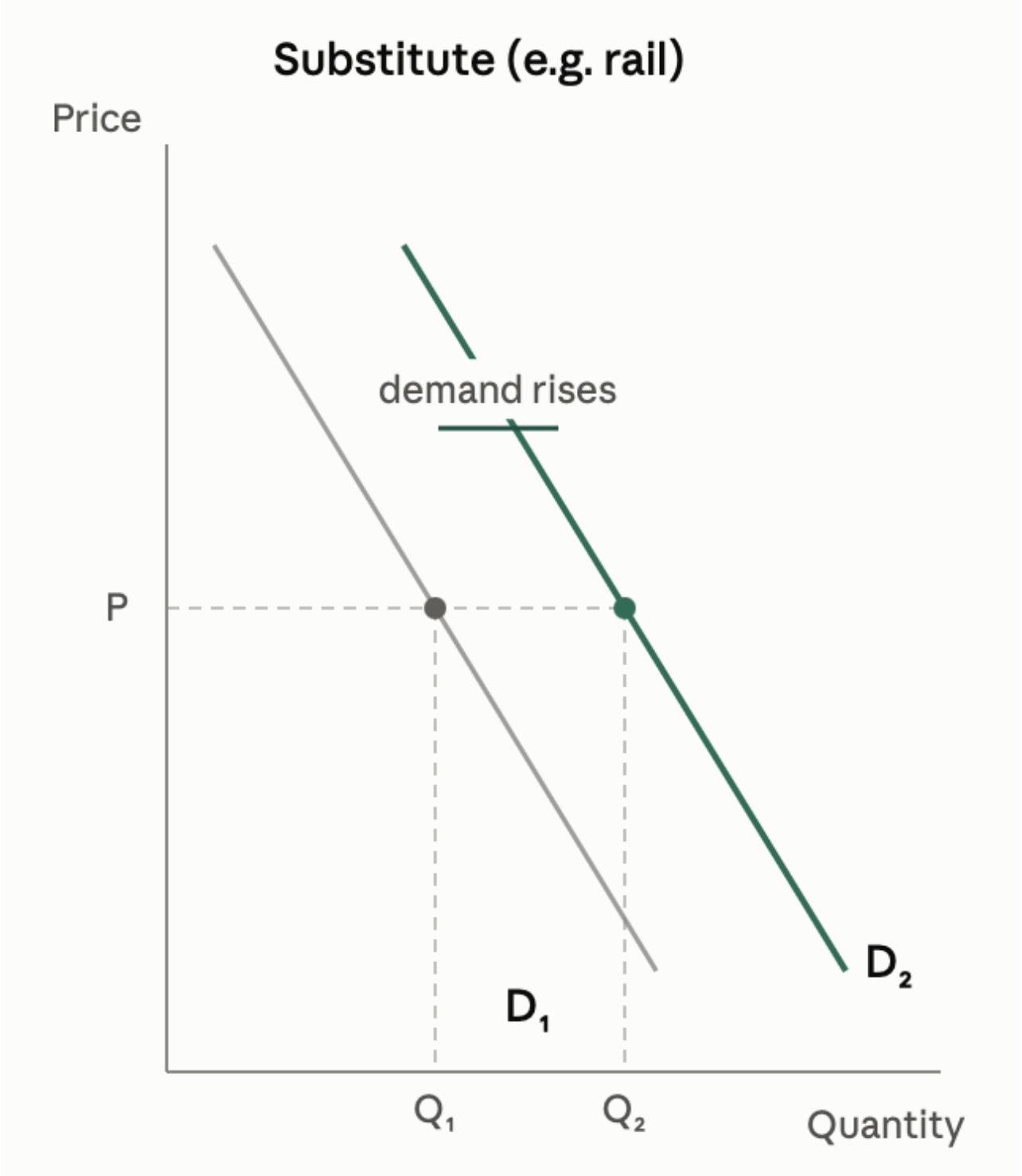
Q16 Explain how an increase in price of air travel might affect the demand for its complements and its substitutes. (10) (2017 SL)
Ans Demand refers to the quantity of a good that consumers are willing and able to purchase at each possible price over a given period of time. The price of related goods is one of the non-price determinants of demand, so a change in the price of one good can shift the entire demand curve of another good that is related to it. Goods can be related as either complements or substitutes, and the two respond in opposite ways to a rise in the price of air travel.

A **complement** is a good that is consumed together with, or jointly with, another good, so that the two tend to be bought and used in combination. Complements for air travel include airport hotel accommodation, holiday packages, car hire at the destination, and travel insurance. When the price of air travel rises, fewer people are willing and able to fly, since flying has become more expensive. Because these complementary goods are consumed jointly with flights, the fall in the quantity of flights purchased reduces the quantity of complements demanded as well.



This is shown in the diagram above. The vertical axis measures the price of the complement (for example, hotel rooms) and the horizontal axis measures the quantity demanded. At the constant price P, the market is initially at the point on demand curve D■, giving quantity Q■. The rise in the price of air travel shifts the whole demand curve for the complement to the left, from D■ to D■, so that at the same price P the relevant point moves to D■ and quantity demanded falls to the smaller Q■. It is important that this is a shift of the curve, not a movement along it, because the change is caused by a non-price factor (the price of a related good) rather than by a change in the complement's own price — the price P is unchanged throughout.

A **substitute** is a good that can be used in place of another to satisfy a similar want. Substitutes for air travel include long-distance rail, coach travel, and — for some journeys — car travel or even video-conferencing in place of a business trip. When the price of air travel rises, some consumers switch away from flying towards these cheaper alternatives, so the quantity of the substitute demanded increases at every price.



This is shown by a **rightward shift** of the demand curve for the substitute, from D■ to D■ — as illustrated in the right-hand panel of the diagram below. For example, a rise in airfares on a domestic route may lead more travellers to take the train instead, raising demand for rail services even though rail fares are unchanged.

It is worth noting that the size of these effects depends on how close the relationship between the goods is. Rail travel is a fairly close substitute for air travel on short domestic routes, so the increase in its demand may be large; but for long intercontinental journeys there is no realistic substitute for flying, so the effect on rail demand would be small or negligible. Similarly, the strength of the complement effect depends on how essential the complement is to air travel — accommodation is closely tied to flying to a foreign destination, whereas travel insurance may be tied more loosely.

In conclusion, a rise in the price of air travel reduces demand for its complements, shifting their demand curves to the left because they are jointly consumed with flights, while it increases demand for its substitutes, shifting their demand curves to the right because consumers switch towards them. In both cases it is a change in a non-price determinant — the price of a related good — that shifts the whole demand curve, and the size of each effect depends on the closeness of the relationship between the goods.

Q17 Explain why products may have different income elasticities of demand. (10) (2023 May HL)
Ans Income elasticity of demand (YED) measures the responsiveness of the quantity demanded of a good to a change in consumer income. It is calculated as the percentage change in quantity demanded divided by the percentage change in income. The value of YED differs from product to product for two main reasons: the *sign* of the YED depends on whether the good is normal or inferior, and the *size* of the YED depends on how large a role the good plays in consumers' spending as their incomes rise.

The first reason products differ is that they fall into two categories with **opposite signs of YED**. A good with a positive YED is a normal good: an increase in income leads to an increase in demand, and a decrease in income leads to a decrease in demand, so quantity demanded moves in the same direction as income. Most goods, such as clothing and cars, are normal. By contrast, a good with a negative YED is an inferior good: an increase in income leads to a fall in demand, because as consumers become richer they switch away from the good towards a higher-quality alternative. Examples include instant noodles or bus travel, which consumers buy less of as their incomes rise. The reason a product has a positive or negative YED therefore lies in how consumers regard it as their incomes change — as something to buy more of, or something to abandon.

The second reason is that, among normal goods, products differ in the **size of their YED**, which splits them into necessities and luxuries. A necessity has a YED between 0 and 1, meaning it is income inelastic: demand rises as income rises, but by proportionally less than the rise in income, because consumers already buy roughly the amount they need of essentials such as staple food. A luxury has a YED greater than 1, meaning it is income elastic: demand rises proportionally more than income, because goods such as foreign holidays or designer clothing are the kinds of purchases consumers expand rapidly once their basic needs are met. The size of the YED thus reflects how far demand for a good expands as income grows once necessities are already satisfied.

These differences can be illustrated with an **Engel curve**, which plots quantity demanded against income, as shown in the diagram below. The curve follows a single good as consumer income rises. Between points A and B the curve is relatively flat, so quantity demanded rises more than

proportionately to income and the good behaves as a luxury (YED greater than 1). Between B and C the curve steepens, with quantity rising less than proportionately to income — the good is now a necessity (0 less than YED less than 1). Between C and D the curve is vertical: quantity demanded stays constant as income rises, so YED equals 0. Beyond D, from D to E, the curve bends backward — quantity demanded actually falls as income rises further — so the good has become an inferior good (YED less than 0). The dashed ray from the origin marks unit income elasticity (YED = 1) as a reference.

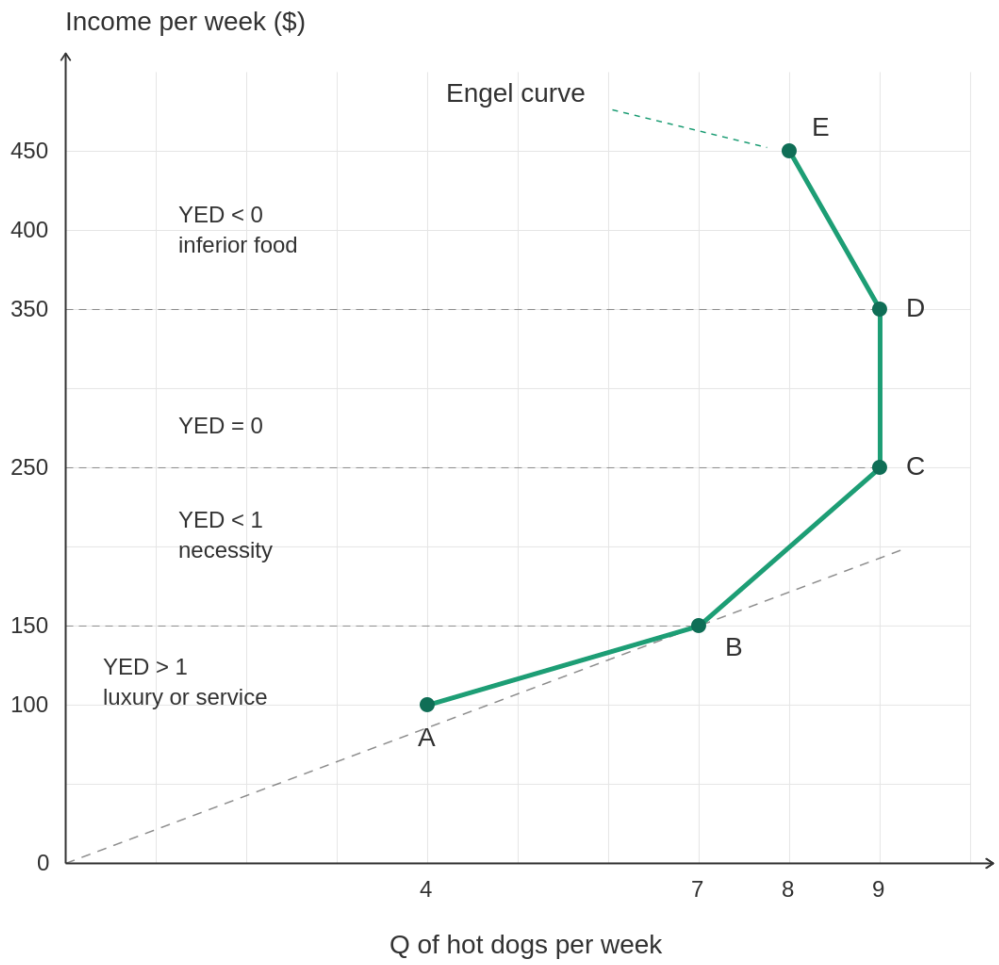


Figure 1 — An Engel curve for a single good, showing how its income elasticity changes as income rises (A–E).

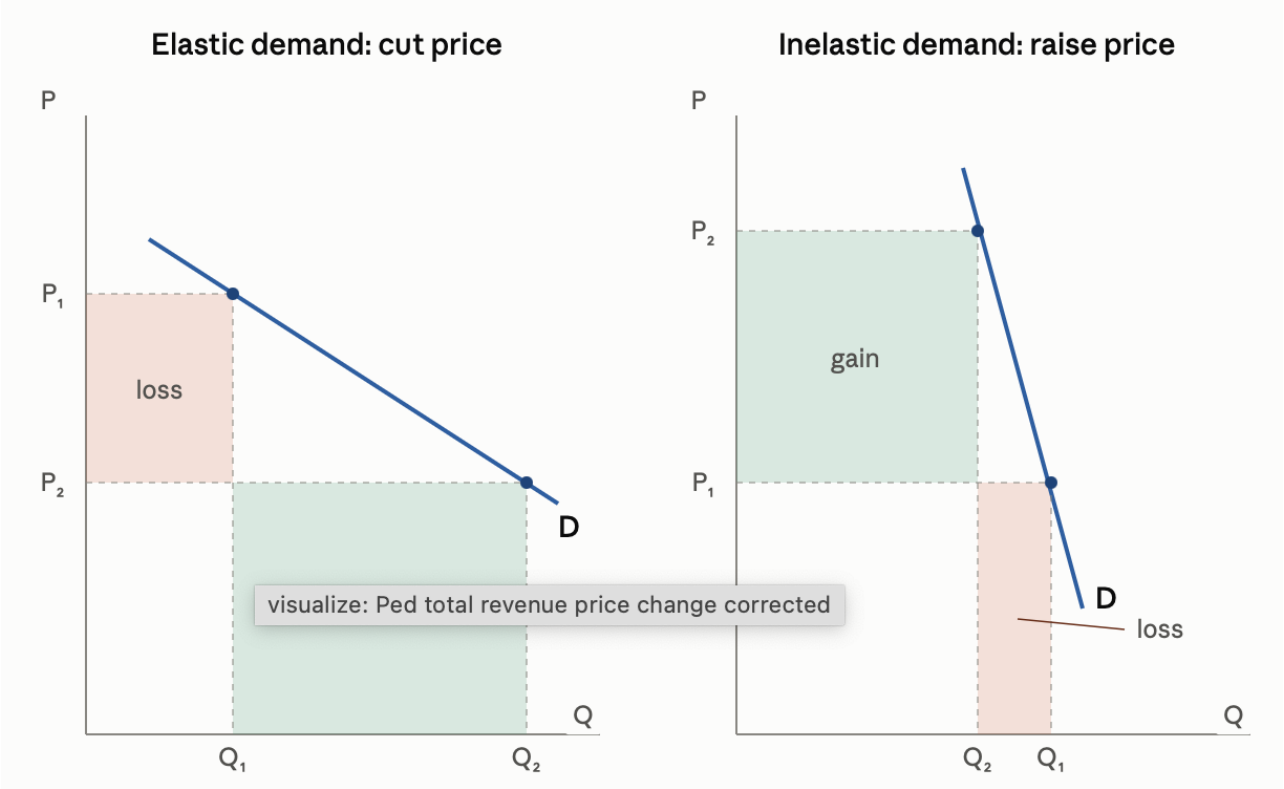
A key insight from the diagram is that the same product can move through all of these categories depending on the consumer's income level, which is precisely why products display different income elasticities.

In conclusion, products have different income elasticities of demand because they play different roles in consumption as incomes change. The sign of the YED differs because normal goods are wanted more as income rises while inferior goods are wanted less, and the size of the YED differs because necessities absorb little extra income once needs are met whereas luxuries absorb a great deal. The particular YED of any product therefore depends on how consumers choose to

spend additional income on it.

Q18 Explain how knowledge of price elasticity of demand could be used by a firm that is considering changing the price of its product. (10) (2020 Nov HL)
Ans Price elasticity of demand (PED) measures the responsiveness of the quantity demanded of a good to a change in its own price, calculated as the percentage change in quantity demanded divided by the percentage change in price. Demand is price elastic when PED is greater than 1 (quantity changes proportionally more than price) and price inelastic when PED is between 0 and 1 (quantity changes proportionally less than price). A firm considering a price change can use knowledge of its product's PED to predict the effect on the quantity sold and, crucially, on its **total revenue** (price × quantity), which is the key link between PED and pricing decisions.

The central use of PED is that it predicts what happens to total revenue when price changes, and the effect runs in opposite directions depending on elasticity, as the diagram below shows.



If a firm knows its product has **price inelastic demand** (PED < 1), quantity is relatively unresponsive to price, so **raising price increases total revenue** — the proportional fall in quantity is smaller than the proportional rise in price. This is the right-hand panel above: raising price from P■ to P■ earns a large "gain" area on the units still sold and loses only the thin "loss" sliver of units given up. A clear real-world example is the US firm Mylan, which raised the price of the EpiPen (a life-saving allergy device with no close substitute) from around $100 to over $600 for a two-pack; because demand was highly inelastic, revenue rose despite the hikes. The same logic explains why tobacco firms can raise cigarette prices with little loss of sales, since nicotine addiction makes demand inelastic.

If instead the firm knows its product has **price elastic demand** (PED > 1), quantity is highly responsive to price, so **lowering price increases total revenue** — the proportional rise in quantity outweighs the proportional fall in price. This is the left-hand panel: cutting price from P■ to P■ loses only the thin top "loss" strip on existing units but gains the large "gain" area from all the extra units sold. A real-world example is budget airlines such as Ryanair, which deliberately set very low fares because air travel between competing carriers is highly price elastic; the low prices fill far more seats, raising total revenue. Similarly, a single brand such as Pepsi faces elastic demand because Coca-Cola is a close substitute, so a price cut can attract many switchers.

Knowledge of PED therefore lets a firm **change price in the correct direction** for its revenue objective, rather than guessing: raise price where demand is inelastic, lower price where demand is elastic. It can also inform related decisions — a firm may use penetration pricing (a low launch price) where it expects elastic demand, as Disney+ did on launch, or spend on advertising and branding to make its own demand more inelastic, as Apple has done, allowing it to charge premium prices for iPhones without losing loyal customers.

There are limits, however. PED estimates are based on past data and may not predict future responses accurately, especially for large price changes; PED also rises over time as consumers find substitutes; and a firm must weigh rivals' reactions, its costs and profit (not just revenue), and the risk that a price rise damages its reputation or attracts new competitors — as Mylan discovered when the EpiPen rises triggered public backlash and political scrutiny.

In conclusion, knowledge of PED is valuable because it predicts the effect of a price change on quantity sold and total revenue. A revenue-seeking firm should raise price when demand is inelastic and lower price when demand is elastic, while treating its PED estimate with caution because it is imprecise and changes over time.